KTU 2024 SCHEME · PCECT302 · S3 ECE

Solid State Devices Playbook

From "why does silicon conduct at all?" to FinFETs, one band diagram at a time.

INSIDE

M1	Semiconductor physics: bands, carriers, drift, diffusion, Hall effect, excess carriers
M2	PN junction, ideal diode equation, capacitance, breakdown and the BJT
M3	Metal–semiconductor contacts, MOS capacitor, threshold voltage, MOSFET I-V
M4	Scaling, subthreshold, short-channel effects, MESFET and FinFET
+	Placement & GATE rapid-fire · formula sheet · night-before summary

★ How to use this playbook

Every chip you will ever design, test or debug is built from the ideas in this subject. A MOSFET is "just" a MOS capacitor with two PN junctions next to it, and a PN junction is "just" what happens when electrons and holes are allowed to wander. If you understand **where the carriers are, how they move, and what the band diagram looks like**, almost every question in this subject (and in core-electronics interviews) becomes a reasoning exercise rather than a memory test.

Each topic follows the same rhythm: **plain-words intuition first** (written for someone who has never seen it), then the **proper theory and derivation** at KTU end-sem depth, then **solved examples** that get gradually harder, and finally an **interview corner**. Learn to sketch the band diagrams yourself; examiners and interviewers both love asking for them.

Suggested routine: one or two sections a day, try each example before reading its solution, and use the revision page at the end of each module the night before a test. Every numerical answer here has been checked with a short Python calculation.

The boxes you will see

Key idea: the one thing to remember

Think of it like: an everyday picture

Exam tip: how marks are won

Common mistake: how marks are lost

Interview corner: placement-style Q&A

Formula: results worth memorising

Constants used throughout (silicon, T = 300 K)

Quantity	Value used	Quantity	Value used
Electron charge q	1.602×10^{-19} C	Thermal voltage $kT/q = V_T$	0.0259 V
Intrinsic concentration n_i	1.5×10^{10} cm^{-3}	Bandgap E_g	1.12 eV
ε_0	8.854×10^{-14} F/cm	$\varepsilon_{Si} = 11.7\varepsilon_0$	1.036×10^{-12} F/cm
$\varepsilon_{ox} = 3.9\varepsilon_0$	3.453×10^{-13} F/cm	N_c, N_v	2.8×10^{19}, 1.04×10^{19} cm^{-3}
μ_n, μ_p (lightly doped)	1350, 480 cm²/V·s	Electron affinity χ_{Si}	4.01 eV

Units: semiconductor work traditionally uses cm, not m. Keep everything in cm (and convert μm → cm by $\times 10^{-4}$, nm → cm by $\times 10^{-7}$) and most unit errors vanish. The textbook pair N_c, N_v gives $n_i \approx 7 \times 10^9$; KTU and most texts still quote $n_i = 1.5 \times 10^{10}$, and so does this book.

≡ Contents

Module 1: Semiconductor Physics & Carrier Transport

1.1 Energy bands: why silicon is a semiconductor 6
1.2 Intrinsic and extrinsic semiconductors 7
1.3 The Fermi–Dirac distribution 8
1.4 Density of states and equilibrium carrier concentration 9
1.5 Fermi level position vs doping and temperature 11
1.6 Drift, mobility and conductivity 13
1.7 The Hall effect 14
1.8 Diffusion and the Einstein relation 16
1.9 Excess carriers: generation, recombination, lifetime 17
1.10 Poisson's equation and the continuity equation 19
1.11 Steady-state solution: the diffusion length 20
1.12 Quasi-Fermi levels 20
Module 1: Quick revision & practice 22

Module 2: PN Junction & BJT

2.1 The PN junction in equilibrium 24
2.2 Depletion width, field and potential profiles 25
2.3 Forward and reverse bias 26
2.4 Derivation of the ideal diode equation 27
2.5 Junction and diffusion capacitance 30
2.6 Junction breakdown: Zener and avalanche 31
2.7 BJT structure and current components 32
2.8 Injection efficiency, transport factor, α and β 33
2.9 Minority carrier distributions in a BJT 35
2.10 The Early effect 36
Module 2: Quick revision & practice 37

Module 3: Metal–Semiconductor Contacts, MOS & MOSFET

3.1 The Schottky barrier 39
3.2 Ohmic contacts 40
3.3 Schottky diode vs PN diode 41
3.4 MOS capacitor: band diagrams 42
3.5 C-V characteristics 43
3.6 Flat-band voltage 44
3.7 Threshold voltage derivation 44

3.8 The body effect 45
3.9 MOSFET structure and operation 47
3.10 Derivation of the MOSFET I-V characteristics 47
3.11 Transconductance 49
Module 3: Quick revision & practice 50

Module 4: Scaling, Short-Channel Effects, MESFET & FinFET

4.1 MOSFET scaling 52
4.2 Subthreshold conduction and subthreshold swing 53
4.3 Short-channel effects and channel length modulation 54
4.4 Drain-induced barrier lowering 55
4.5 Velocity saturation 56
4.6 Hot-carrier effects 57
4.7 Threshold voltage roll-off 57
4.8 The MESFET 59
4.9 The FinFET 60
Module 4: Quick revision & practice 61

End matter

Placement & GATE rapid-fire 62
Formula sheet 64
Night before the exam 66

01

Semiconductor Physics & Carrier Transport

> *Every device in this course (diode, BJT, MOSFET, FinFET) is explained with the carrier ideas built here. Interviewers start with "draw the band diagram" for a reason.*

SYLLABUS CHECKLIST

- ☐ Energy bands; intrinsic and extrinsic semiconductors
- ☐ Fermi–Dirac distribution; density of states
- ☐ Equilibrium carrier concentration: n_0, p_0, n_i, mass-action law
- ☐ Fermi level position vs doping and temperature
- ☐ Carrier transport: drift, mobility, conductivity
- ☐ Hall effect: Hall voltage, carrier type, concentration and mobility
- ☐ Diffusion; Einstein relation
- ☐ Excess carriers: generation, recombination, lifetime
- ☐ Poisson's equation; continuity equation; steady-state diffusion length solution
- ☐ Quasi-Fermi levels

13 hours · CO1 · Explain carrier statistics and transport in semiconductors

1.1 Energy bands: why silicon is a semiconductor

An isolated atom has electrons sitting at sharp, separate energy levels, like rungs on a ladder. Now bring 10^{22} silicon atoms together to form a crystal. Their outer electrons start to "feel" each other, and each sharp level spreads into a **band** of extremely closely spaced levels. Two bands matter for us:

- **Valence band** (top edge E_V): the electrons that form the covalent bonds. At 0 K it is completely full.
- **Conduction band** (bottom edge E_C): empty at 0 K. An electron here is free to move through the crystal and carry current.

Between them lies the **bandgap** $E_g = E_C - E_V$, a range of energies where no electron states exist in a perfect crystal. For Si E_g = 1.12 eV, for GaAs 1.42 eV, for Ge 0.66 eV.

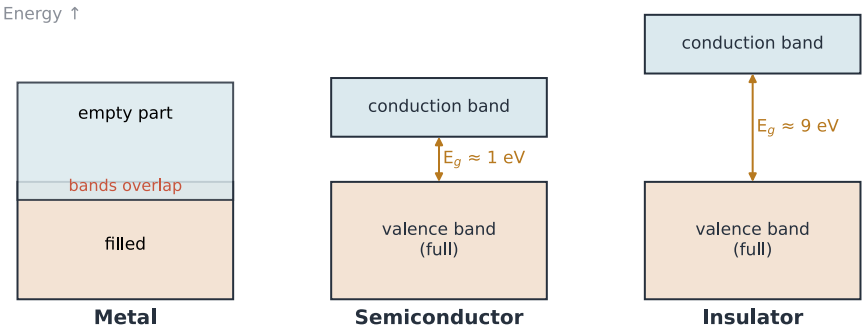


Fig 1.1 Metal, semiconductor and insulator differ only in how big the gap is (or whether there is one at all).

THINK OF IT LIKE

A two-storey car park. The ground floor (valence band) is packed bumper to bumper: no car can move. The upper floor (conduction band) is empty. Lift one car upstairs (give an electron ≥ E_g of energy) and *two* things can now move: the car upstairs, and the empty slot downstairs, which shuffles along as neighbouring cars move into it. That moving empty slot is a **hole**.

So a semiconductor conducts with **two carriers**: electrons in the conduction band (charge −q) and holes in the valence band (charge +q). The gap is small enough (about 1 eV) that heat at room temperature excites a few electrons across it, so conduction is small but non-zero, and very sensitive to temperature, light and impurities. That sensitivity is exactly what makes devices possible.

KEY IDEA

Metal: bands overlap, so there are always free electrons. Insulator: gap so large (SiO_2 ≈ 9 eV) that essentially no electron crosses. Semiconductor: gap of about 1 eV, so a controllable number crosses. Holes are not "imaginary"; they behave as real positive particles with their own effective mass and mobility.

Effective mass. Inside a crystal an electron is pushed around by the periodic lattice potential as well as by the applied field. We hide all that in one number, the effective mass m*: the electron responds to an external force as if its mass were m*. For Si, $m_n^* \approx 1.08\ m_0$ (density-of-states value) and $m_p^* \approx 0.56\ m_0$.

Direct vs indirect gap. In GaAs the bottom of the conduction band sits right above the top of the valence band in momentum (k), so an electron can drop down and emit a photon directly: a **direct** gap, good for LEDs and lasers. In Si the minimum is shifted in k, so a transition also needs a phonon (lattice vibration). Si is **indirect**, which is why it is a poor light emitter.

1.2 Intrinsic and extrinsic semiconductors

An **intrinsic** semiconductor is perfectly pure. Every electron that jumps to the conduction band leaves exactly one hole behind, so $n = p = n_i$, the intrinsic concentration (1.5×10^{10} cm^{-3} for Si at 300 K). That is about one free electron per 3×10^{12} atoms: almost an insulator.

We make silicon useful by **doping**, which means adding a tiny, controlled amount of impurity:

- **Donors** (group V: P, As, Sb) have 5 outer electrons. Four bond with neighbours; the fifth is loosely held, only about 0.045 eV below E_C (the donor level E_d). At room temperature it is set free almost always. Result: an **n-type** material with $n \gg p$ and fixed **positive** donor ions left behind.
- **Acceptors** (group III: B, Al, Ga) have only 3 outer electrons. They grab an electron from the valence band (acceptor level E_a just above E_V), creating a hole. Result: **p-type**, $p \gg n$, with fixed **negative** acceptor ions.

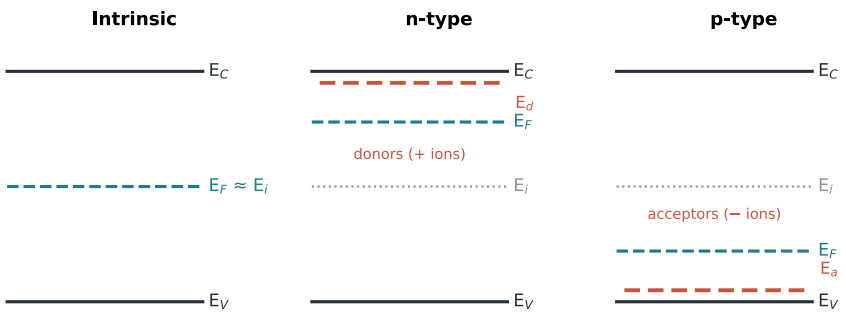


Fig 1.2 Band diagrams: doping moves the Fermi level E_F towards E_C (n-type) or towards E_V (p-type).

In n-type material electrons are the **majority carriers** and holes the **minority carriers**; in p-type it is the other way round. Even 10^{16} cm^{-3} donors (one impurity per 5 million Si atoms) raise the electron count by a factor of almost a million. If a sample contains both donors and acceptors it is **compensated**: they cancel, and the net doping $|N_d - N_a|$ decides the type.

COMMON MISTAKE

"n-type material is negatively charged." No: every free electron came from a neutral donor atom that is now a + ion. The crystal as a whole stays **electrically neutral**. Charge neutrality, $n_0 + N_a^- = p_0 + N_d^+$, is one of the two equations used for every concentration calculation.

1.3 The Fermi–Dirac distribution

Electrons obey the Pauli exclusion principle: at most one electron per quantum state. The probability that a state at energy E is occupied, in thermal equilibrium, is the **Fermi–Dirac function**:

FERMI–DIRAC FUNCTION

$$f(E) = \frac{1}{1 + \exp[(E - E_F)/kT]}$$

Probability a state is **empty** (a hole, if in the valence band) = $1 - f(E)$.

E_F, the Fermi level, is the energy at which $f = ½$. At 0 K the function is a step: every state below E_F is full and every state above is empty. As temperature rises, the step softens over a few kT on either side (kT = 0.0259 eV at 300 K).

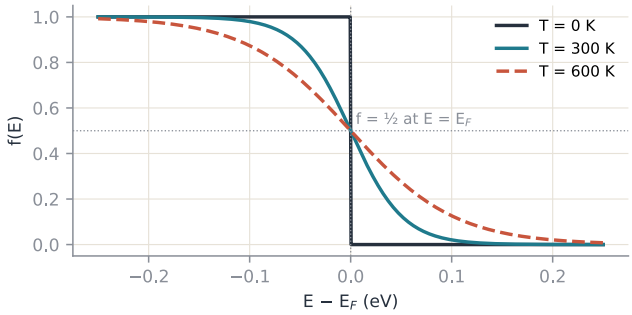


Fig 1.3 f(E) at 0, 300 and 600 K. It is always ½ at E_F, and the curve is symmetric about that point.

Boltzmann approximation. If $E - E_F > 3kT$, the "1" in the denominator is small and $f(E) \approx \exp[-(E - E_F)/kT]$. This is the classical (Maxwell–Boltzmann) limit, and it is what we use for non-degenerate semiconductors, where E_F sits inside the gap at least 3kT from either band edge.

SOLVED EXAMPLE 1.1 · FERMI FUNCTION

(a) Find the probability that a state 3kT above E_F is occupied, exactly and with the Boltzmann approximation. (b) At 300 K, find the probability that a state 0.1 eV above E_F is occupied and that a state 0.1 eV below E_F is empty.

▸ (a) Exact: $f = 1/(1 + e^{3}) = 1/21.09 = 0.0474$. Boltzmann: $f \approx e^{-3} = 0.0498$, an error of about 5 %, which is why "3kT" is the usual validity limit.

▸ (b) $(E - E_F)/kT = 0.1/0.0259 = 3.86$. $f = 1/(1 + e^{3.86}) = 0.0206$.

▸ Empty probability 0.1 eV below: $1 - f = 1 - 1/(1 + e^{-3.86}) = 0.0206$, the same number, by the symmetry of f about E_F.

(a) 0.0474 (exact), 0.0498 (Boltzmann) · (b) 0.0206 in both cases

1.4 Density of states and equilibrium carrier concentration

To count electrons in the conduction band we need two things: **how many seats exist** at each energy, and **what fraction is occupied**. The first is the **density of states** g(E) (states per unit volume per unit energy); the second is f(E).

DENSITY OF STATES (3-D, PARABOLIC BAND)

$$g_c(E) = \frac{4\pi(2m_n^*)^{3/2}}{h^3}\sqrt{(E - E_C)} \quad (E \geq E_C) \qquad g_v(E) = \frac{4\pi(2m_p^*)^{3/2}}{h^3}\sqrt{(E_V - E)}$$

It grows like a square root as you go deeper into either band. Multiplying the number of seats by the occupancy gives the electron distribution $n(E) = g_c(E)\, f(E)$; holes are $p(E) = g_v(E)[1 - f(E)]$. The area under each curve is the total concentration.

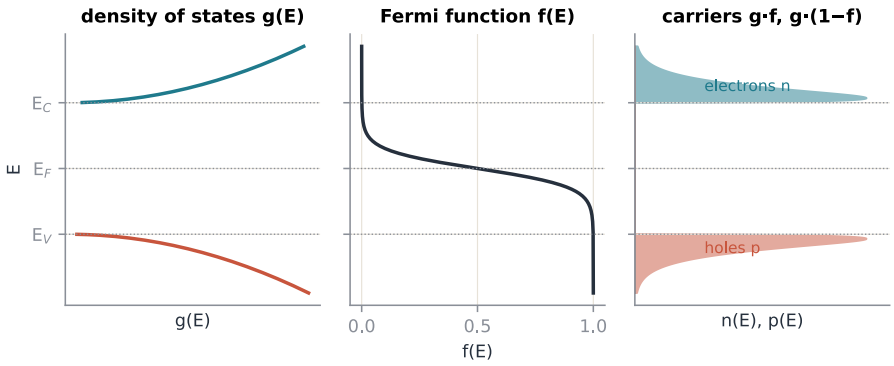


Fig 1.4 Seats × occupancy = carriers (intrinsic case, kT exaggerated for visibility). Carriers crowd within a few kT of each band edge.

Deriving n_0 and p_0

$n_0 = \int_{E_C}^{\infty} g_c(E)\, f(E)\, dE$. With the Boltzmann approximation $f \approx \exp[-(E - E_F)/kT]$.

Substitute $\eta = (E - E_C)/kT$: $n_0 = \frac{4\pi(2m_n^* kT)^{3/2}}{h^3} \exp[-(E_C - E_F)/kT] \int_0^{\infty} \eta^{1/2} e^{-\eta}\, d\eta$.

The integral is the gamma function $\Gamma(3/2) = \sqrt{\pi}/2$. Collecting constants gives the **effective density of states** $N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$.

The same steps for holes give p_0 with $N_v = 2(2\pi m_p^* kT/h^2)^{3/2}$.

EQUILIBRIUM CONCENTRATIONS

$$n_0 = N_c \exp[-(E_C - E_F)/kT] \qquad p_0 = N_v \exp[-(E_F - E_V)/kT]$$

$$n_0\, p_0 = n_i^2 = N_c N_v \exp(-E_g/kT) \quad \text{(mass-action law)}$$

$$n_0 = n_i\, e^{(E_F - E_i)/kT} \qquad p_0 = n_i\, e^{(E_i - E_F)/kT}$$

The **mass-action law** is the star result: the product $n_0 p_0$ does not depend on E_F, so it does not depend on doping. Add donors and electrons go up, but holes go down by the same factor (the extra electrons fall into holes and annihilate them).

General formula. Combine charge neutrality $n_0 + N_a = p_0 + N_d$ (full ionisation) with $n_0 p_0 = n_i^2$ and solve the quadratic:

EXACT SOLUTION (ANY DOPING, ANY TEMPERATURE)

$$n_0 = \frac{N_d - N_a}{2} + \sqrt{\left[\left((N_d - N_a)/2\right)^2 + n_i^2\right]} \qquad p_0 = n_i^2/n_0$$

If $N_d - N_a \gg n_i$: $n_0 \approx N_d - N_a$. This is almost always true at room temperature.

SOLVED EXAMPLE 1.2 · N-TYPE SILICON

Si at 300 K is doped with $N_d = 10^{16}$ cm^{-3}. Find n_0, p_0, the position of E_F relative to E_i, and relative to E_C.

- $N_d \gg n_i$, so $n_0 = 10^{16}$ cm^{-3}.
- $p_0 = n_i^2/n_0 = (1.5 \times 10^{10})^2/10^{16} = 2.25 \times 10^4$ cm^{-3}.
- $E_F - E_i = kT \ln(n_0/n_i) = 0.0259 \times \ln(6.67 \times 10^5) = 0.347$ eV.
- $E_C - E_F = kT \ln(N_c/n_0) = 0.0259 \times \ln(2800) = 0.206$ eV.

$n_0 = 10^{16}$, $p_0 = 2.25 \times 10^4$ cm^{-3}; E_F is 0.347 eV above E_i and 0.206 eV below E_C

The two answers do not add up to exactly $E_g/2 = 0.56$ eV because the textbook N_c and $n_i = 1.5 \times 10^{10}$ are slightly inconsistent (see the constants note at the front). Use whichever form the question supplies data for.

SOLVED EXAMPLE 1.3 · COMPENSATED SILICON

A Si sample has $N_a = 5 \times 10^{16}$ and $N_d = 2 \times 10^{16}$ cm^{-3}. Is it n- or p-type? Find n_0, p_0 and $E_i - E_F$.

- $N_a > N_d \Rightarrow$ p-type. $p_0 \approx N_a - N_d = 3 \times 10^{16}$ cm^{-3}.
- $n_0 = n_i^2/p_0 = 2.25 \times 10^{20}/3 \times 10^{16} = 7.5 \times 10^3$ cm^{-3}.
- $E_i - E_F = kT \ln(p_0/n_i) = 0.0259 \times \ln(2 \times 10^6) = 0.376$ eV.

p-type; $p_0 = 3 \times 10^{16}$, $n_0 = 7.5 \times 10^3$ cm^{-3}; E_F is 0.376 eV below E_i

1.5 Fermi level position vs doping and temperature

Intrinsic Fermi level

Setting $n_0 = p_0$ in the two exponential formulas gives

INTRINSIC FERMI LEVEL

$$E_{Fi} - E_{midgap} = \frac{3}{4} kT \ln(m_p^*/m_n^*)$$

If the two effective masses were equal, E_i would sit exactly mid-gap. They are not equal, so E_i is shifted a tiny bit towards the band with the *lighter* carriers (fewer states, so E_F must move closer to it to keep n = p). For Si ($m_n^* = 1.08m_0$, $m_p^* = 0.56m_0$) the shift is $0.75 \times 0.0259 \times \ln(0.56/1.08) = -12.8$ meV: about 1 % of E_g, which is why we treat E_i as mid-gap.

Effect of doping and temperature

From $n_0 = n_i\, e^{(E_F - E_i)/kT}$:

- **Doping:** $E_F - E_i = kT \ln(N_d/n_i)$ for n-type, and $E_i - E_F = kT \ln(N_a/n_i)$ for p-type. Every 10× increase in doping moves E_F by $kT \ln 10 = 59.6$ meV towards the band edge. At very high doping (around N_c) E_F reaches the band edge and the material is **degenerate** (behaves like a metal; Boltzmann formulas fail).
- **Temperature:** as T rises n_i grows exponentially, $n_i \propto T^{3/2} \exp(-E_g/2kT)$. Once n_i approaches the doping, the material behaves as intrinsic and E_F drifts back to E_i. Heavily doped samples hold out to higher temperatures.

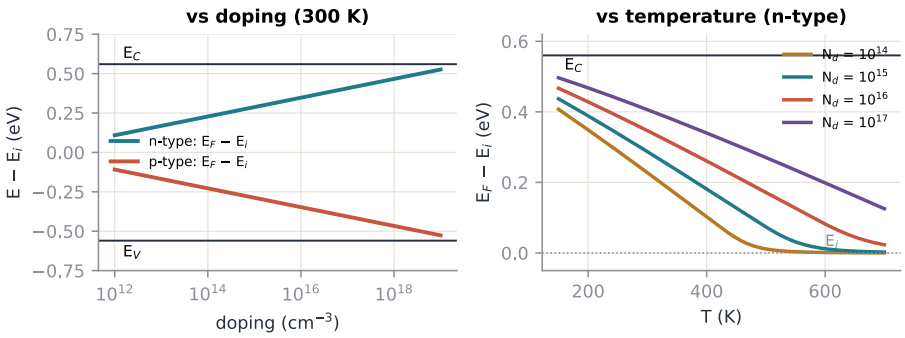


Fig 1.5 Left: E_F moves 59.6 meV per decade of doping. Right: with rising T every sample eventually turns intrinsic ($E_F \to E_i$); heavier doping survives longer.

Three temperature regions (Fig 1.6). At very low T even donors cannot ionise: **freeze-out**, $n < N_d$. Over a wide middle range (roughly 100 K to 450 K for 10^{15} cm^{-3}) all donors are ionised and $n \approx N_d$: the **extrinsic** region, where devices are designed to work. At high T, n_i overtakes N_d: the **intrinsic** region, where the device loses its doping-defined behaviour. This is why Si chips are rated up to about 125–150 °C, and wide-gap SiC/GaN go much higher.

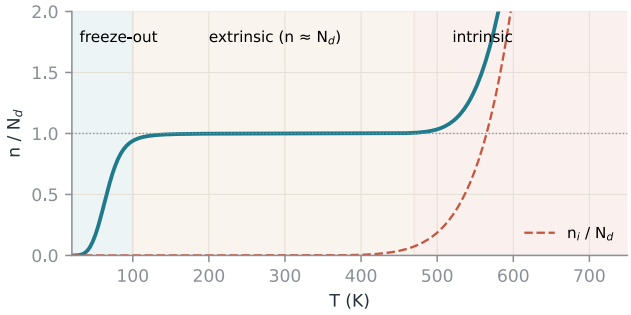


Fig 1.6 Electron concentration vs temperature for $N_d = 10^{15}$ cm^{-3} (computed with donor level 45 meV below E_C).

SOLVED EXAMPLE 1.4 · HOT SILICON GATE

Taking $n_i \propto T^{3/2}\exp(-E_g/2kT)$ with $E_g = 1.12$ eV constant, find n_i at 400 K. A Si sample with $N_d = 10^{15}$ cm^{-3} is heated to 400 K. Find n_0 and p_0.

- Ratio $= (400/300)^{3/2} \times \exp[-(1.12/2k)(1/400 - 1/300)]$ with $k = 8.617 \times 10^{-5}$ eV/K.
- $= 1.540 \times \exp(5.42) = 1.540 \times 225 \approx 346$, so $n_i(400\text{ K}) \approx 1.5 \times 10^{10} \times 346 = 5.19 \times 10^{12}$ cm^{-3}.
- $N_d/2 = 5 \times 10^{14} \gg n_i$, so the exact formula still gives $n_0 \approx 1.00 \times 10^{15}$ cm^{-3} (still extrinsic).
- $p_0 = n_i^2/n_0 = (5.19 \times 10^{12})^2/10^{15} = 2.7 \times 10^{10}$ cm^{-3}, about a million times its 300 K value.

$n_i \approx 5.2 \times 10^{12}$ cm^{-3}; $n_0 \approx 10^{15}$, $p_0 \approx 2.7 \times 10^{10}$ cm^{-3}

EXAM TIP

Minority carriers are the temperature-sensitive ones: majority stays $\approx N_d$, but minority $\propto n_i^2$, which roughly doubles every 5–6 °C near room temperature in Si. That is why a diode's reverse saturation current (made of minority carriers) is so temperature sensitive.

INTERVIEW CORNER

Q. What is the Fermi level, physically?

A. The energy at which the occupancy probability is ½; equivalently, the electrochemical potential of electrons. In equilibrium it is flat (constant) throughout any connected system, which is the rule behind every band diagram you draw.

Q. Why does adding donors reduce the hole concentration?

A. Mass-action: $np = n_i^2$. More electrons means more recombination with holes, so the equilibrium hole count drops by the same factor.

Q. What is a degenerate semiconductor?

A. Doping so high ($\gtrsim N_c$) that E_F enters the band. Boltzmann statistics no longer apply, impurity levels merge into a band and the bandgap narrows. Used in tunnel diodes and in ohmic contact regions.

1.6 Drift, mobility and conductivity

Even with no field, carriers race around at the thermal velocity ($\approx 10^7$ cm/s), but in random directions, constantly bouncing off vibrating atoms and impurity ions. Net motion: zero. Apply a field E and each carrier gets a small push between collisions, giving a slow average **drift velocity** superimposed on the random motion:

DRIFT

$$v_{dp} = \mu_p E \qquad v_{dn} = -\mu_n E \qquad \mu = \frac{q\tau_c}{m^*}$$

$$J_{drift} = q(n\mu_n + p\mu_p)E = \sigma E \qquad \sigma = q(n\mu_n + p\mu_p), \quad \rho = 1/\sigma$$

Here τ_c is the mean time between collisions. Electrons and holes drift in opposite directions, but because their charges are opposite, their **currents add**. Electrons are more mobile than holes in Si (1350 vs 480 $cm^2/V{\cdot}s$), which is why NMOS is faster than PMOS of the same size.

What limits mobility? Two scattering mechanisms. **Lattice (phonon) scattering** grows with temperature, so μ falls as T rises (roughly $T^{-3/2}$). **Ionised impurity scattering** grows with doping and is weaker at high T (fast carriers fly past the ions). The combined mobility follows Matthiessen's rule: $1/\mu = 1/\mu_L + 1/\mu_I$. At high fields the drift velocity stops growing and saturates at about 10^7 cm/s, a fact that will matter a lot in Module 4.

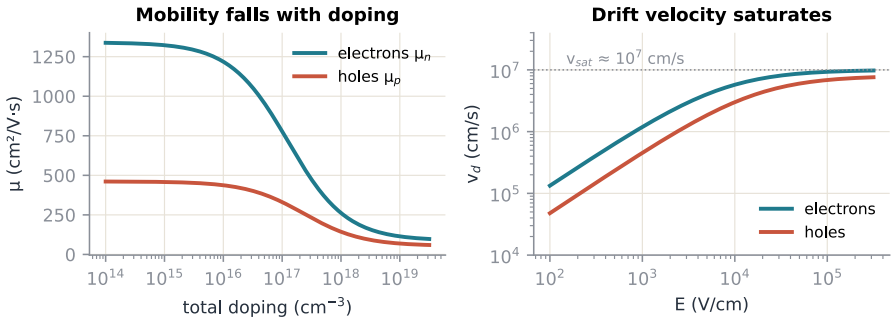


Fig 1.7 Left: mobility falls with doping (impurity scattering). Right: drift velocity is linear in E at low field and saturates at high field.

SOLVED EXAMPLE 1.5 · CONDUCTIVITY AND DRIFT

Si doped with $N_d = 10^{16}$ cm^{-3}, $\mu_n = 1350$ cm²/V·s. Find σ, ρ, the drift current density and the electron drift velocity for E = 10 V/cm. Compare ρ with intrinsic Si ($\mu_p = 480$).

▸ Holes are negligible (2.25×10^4). $\sigma = qn\mu_n = 1.602 \times 10^{-19} \times 10^{16} \times 1350 = 2.16$ S/cm, so $\rho = 0.462$ Ω·cm.

▸ $J = \sigma E = 2.16 \times 10 = 21.6$ A/cm². $|v_d| = \mu_n E = 1.35 \times 10^4$ cm/s (far below v_{th}, so the linear model holds).

▸ Intrinsic: $\rho_i = 1/[qn_i(\mu_n + \mu_p)] = 1/(1.602 \times 10^{-19} \times 1.5 \times 10^{10} \times 1830) = 2.27 \times 10^5$ Ω·cm.

σ = 2.16 S/cm, ρ = 0.462 Ω·cm, J = 21.6 A/cm², $v_d = 1.35 \times 10^4$ cm/s; doping cut ρ by about 5×10^5

1.7 The Hall effect

How do you find out, experimentally, whether a sample is n- or p-type, and how many carriers it has? Pass a current through it and put it in a magnetic field. Moving charges feel the Lorentz force F = q(v × B), which pushes them to one side of the bar. Charge piles up on that side until the resulting transverse electric field (the **Hall field**) exactly cancels the magnetic push. The voltage across the bar is the **Hall voltage** V_H.

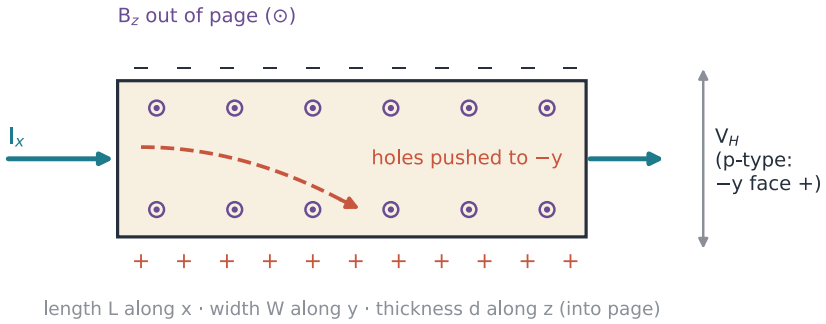


Fig 1.8 Hall set-up seen from above: I_x along x, B_z out of the page. Holes (and electrons, which move the opposite way) are both pushed to the −y face.

The clever part: holes move along +x and electrons along −x, but both get pushed to the **same** face (−y). So that face becomes **positive for p-type** and **negative for n-type**. The **sign of V_H tells you the carrier type**.

Derivation (p-type)

Steady state: no net y-force, so $qE_H = qv_xB_z$, i.e. $E_H = v_xB_z$.
Hall voltage across width W: $V_H = E_HW = v_xB_zW$.
Current $I_x = qpv_x(Wd)$, so $v_x = I_x/(qpWd)$. Substituting: $V_H = I_xB_z/(qpd)$.
Mobility from the ordinary resistance along x: $I_x = qp\mu_p(V_x/L)(Wd)$.

HALL EFFECT

$$V_H = \frac{I_xB_z}{qpd} \quad \Rightarrow \quad p = \frac{I_xB_z}{qdV_H} \qquad \mu_p = \frac{I_xL}{qpV_xWd}$$

$$\text{Hall coefficient } R_H = \frac{1}{qp} \text{ (p-type, } R_H > 0), \quad R_H = -\frac{1}{qn} \text{ (n-type)} \qquad \mu = \sigma|R_H|$$

COMMON MISTAKE

Using the width W instead of the thickness d in $V_H = IB/(qpd)$. The relevant dimension is the one **parallel to B**. Also, work in SI units (m, T) for Hall problems, or convert B: 1 T = 10^4 gauss.

SOLVED EXAMPLE 1.6 · HALL MEASUREMENT FREQUENTLY ASKED

A Si bar has L = 2 mm, W = 0.5 mm, d = 20 µm. With I_x = 2 mA, V_x = 5 V and B_z = 0.1 T, the Hall voltage is +4 mV (−y face positive). Find the carrier type, concentration, mobility and R_H.

▸ V_H positive on the −y face ⇒ **p-type**.

▸ $p = I_xB_z/(qdV_H) = (2 \times 10^{-3} \times 0.1)/(1.602 \times 10^{-19} \times 20 \times 10^{-6} \times 4 \times 10^{-3}) = 1.56 \times 10^{22}$ m^{-3} = 1.56×10^{16} cm^{-3}.

▸ $\mu_p = I_x L/(qpV_x Wd) = (2 \times 10^{-3} \times 2 \times 10^{-3})/(1.602 \times 10^{-19} \times 1.56 \times 10^{22} \times 5 \times 0.5 \times 10^{-3} \times 20 \times 10^{-6}) = 0.032\ m^2/V{\cdot}s$.

▸ $R_H = 1/(qp) = 4 \times 10^{-4}\ m^3/C$.

p-type, $p = 1.56 \times 10^{16}\ cm^{-3}$, $\mu_p = 320\ cm^2/V{\cdot}s$, $R_H = 4 \times 10^{-4}\ m^3/C$

1.8 Diffusion and the Einstein relation

Drop ink into still water and it spreads out, with no force pushing it. Random motion alone moves particles from where there are many to where there are few. Carriers do the same. A concentration gradient causes a **diffusion current**, proportional to the gradient:

DIFFUSION CURRENT DENSITY

$$J_{n,diff} = +qD_n \frac{dn}{dx} \qquad J_{p,diff} = -qD_p \frac{dp}{dx}$$

Total: $J_n = qn\mu_n E + qD_n\, dn/dx \qquad J_p = qp\mu_p E - qD_p\, dp/dx$

The signs trip people up. Particles always diffuse *down* the gradient. Holes are positive, so hole current flows down the gradient too: hence the minus sign. Electrons are negative, so their current flows opposite to their motion: up the gradient, plus sign.

Einstein relation: D and μ are linked

Both D and μ measure how freely a carrier moves between collisions, so they must be related. Proof: in equilibrium the total current is zero, even in a non-uniformly doped sample.

$0 = qn\mu_n E + qD_n\, dn/dx$. Non-uniform doping means E_i (and so the bands) varies with x while E_F stays flat.
$n = n_i \exp[(E_F - E_i)/kT] \Rightarrow dn/dx = -(n/kT)(dE_i/dx)$.
The field is the slope of the bands: $E = (1/q)(dE_i/dx)$. Substituting gives $n\mu_n(1/q)(dE_i/dx) = D_n(n/kT)(dE_i/dx)$.
Cancel: $D_n/\mu_n = kT/q$. Same for holes.

EINSTEIN RELATION

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q} = V_T = 0.0259\ V \text{ at } 300\ K$$

SOLVED EXAMPLE 1.7 · DIFFUSION CURRENT

(a) Find D_n and D_p in Si for $\mu_n = 1350$ and $\mu_p = 480\ cm^2/V{\cdot}s$. (b) Electron concentration falls linearly from 5×10^{16} to $10^{16}\ cm^{-3}$ over 20 μm (taking $D_n = 35\ cm^2/s$). Find the diffusion current density and its direction.

- (a) $D_n = 0.0259 \times 1350 = 35.0$ cm²/s; $D_p = 0.0259 \times 480 = 12.4$ cm²/s.
- (b) $dn/dx = (10^{16} - 5 \times 10^{16})/(20 \times 10^{-4}) = -2 \times 10^{19}$ cm^{-4}.
- $J_n = qD_n dn/dx = 1.602 \times 10^{-19} \times 35 \times (-2 \times 10^{19}) = -112$ A/cm².
- Electrons diffuse towards +x (downhill); the current is in −x.

$D_n = 35$, $D_p = 12.4$ cm²/s; $J_n = -112$ A/cm² (current along −x)

INTERVIEW CORNER

Q. Can current flow without an electric field?
A. Yes, by diffusion. In a forward-biased diode's neutral regions, the minority current is almost purely diffusion.

Q. Why is $\mu_n > \mu_p$ in silicon?
A. Electrons in the conduction band have a smaller conductivity effective mass ($\approx 0.26m_0$) than holes ($\approx 0.39m_0$), so they accelerate more between collisions.

Q. What does the Hall effect tell you that a resistivity measurement cannot?
A. Resistivity gives only the product nμ. The Hall voltage gives n (and the carrier type) separately, so μ can be extracted.

1.9 Excess carriers: generation, recombination, lifetime

So far everything was in **thermal equilibrium**, where thermal generation of electron-hole pairs exactly balances recombination. Shine light on the sample (photon energy > E_g) or inject carriers from a junction, and you create **excess carriers** $\delta n = \delta p$ above the equilibrium values:

$$n = n_0 + \delta n, \quad p = p_0 + \delta p, \quad \text{and now } np > n_i^2$$

Nature tries to restore equilibrium: excess electrons and holes **recombine**. Three main routes:

- **Direct (band-to-band)**: an electron drops straight into a hole, often emitting a photon. Dominant in direct-gap GaAs.
- **Indirect, via traps (Shockley–Read–Hall)**: a defect level near mid-gap catches an electron, then a hole. Dominant in Si. Gold is deliberately added to fast-switching Si diodes to create such traps and shorten lifetime.
- **Auger**: the energy goes to a third carrier. Important only at very high concentrations.

Low-level injection means $\delta p \ll$ majority concentration (e.g. $\delta p \ll n_0$ in n-type). Then the majority count barely changes and the recombination rate of the excess minority carriers is simply proportional to their number:

RECOMBINATION AND LIFETIME (LOW-LEVEL INJECTION, N-TYPE)

$$R = \frac{\delta p}{\tau_p} \qquad \frac{d\delta p}{dt} = G - \frac{\delta p}{\tau_p}$$

Steady state: $\delta p = G\tau_p$ Light switched off: $\delta p(t) = \delta p(0)\, e^{-t/\tau_p}$

τ_p is the **minority carrier lifetime**: the average time an excess hole survives before recombining. In good Si it ranges from nanoseconds to milliseconds depending on purity. It sets diode switching speed (Module 2) and solar-cell efficiency.

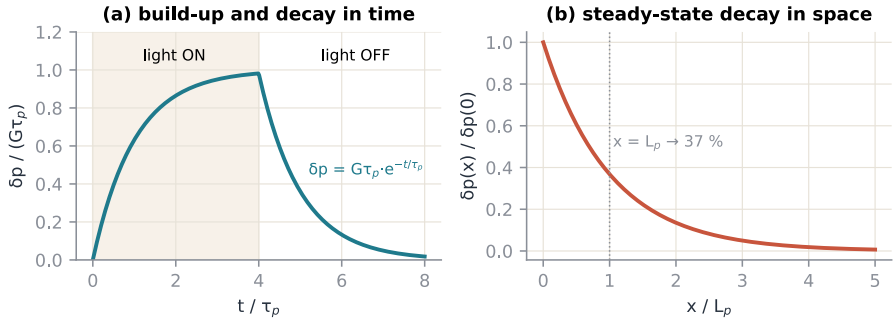


Fig 1.9 (a) Excess holes build up and decay with time constant τ_p. (b) With steady injection at x = 0, they decay in space with the diffusion length L_p (section 1.11).

THINK OF IT LIKE

A water tank with a leak at the bottom (recombination) and a tap at the top (generation). With the tap on, the level rises until the leak rate equals the inflow: $\delta p = G\tau$. Turn off the tap and it drains exponentially.

SOLVED EXAMPLE 1.8 · PHOTOGENERATION

n-type Si ($N_d = 10^{16}$ cm^{-3}, $\tau_p = 1$ µs) is uniformly illuminated with $G = 10^{20}$ cm^{-3}s^{-1}. Find the steady-state δp. The light is switched off; find δp after 2 µs and the time for δp to fall to 1 % of its initial value. Is it low-level injection?

- $\delta p = G\tau_p = 10^{20} \times 10^{-6} = 10^{14}$ cm^{-3}. Since $10^{14} \ll 10^{16}$: yes, low-level injection.
- $\delta p(2\ \mu s) = 10^{14} \times e^{-2} = 1.35 \times 10^{13}$ cm^{-3}.
- $e^{-t/\tau} = 0.01 \Rightarrow t = \tau \ln 100 = 4.6$ µs.

$\delta p = 10^{14}$ cm^{-3}; 1.35×10^{13} cm^{-3} at 2 µs; 4.6 µs to reach 1 %

Notice what happened to the hole concentration: from 2.25×10^4 to 10^{14}, an enormous relative change, while electrons went from 1.00×10^{16} to 1.01×10^{16}. **Excess carriers matter hugely for minority carriers and barely at all for majority carriers.** Almost every device analysis is therefore an analysis of minority carriers.

INTERVIEW CORNER

Q. What is the difference between lifetime and diffusion length?
A. Lifetime is how long an excess minority carrier survives (a time); diffusion length $L = \sqrt{(D\tau)}$ is how far it wanders in that time (a distance).

Q. Why is Si a poor LED material?
A. It has an indirect gap: radiative recombination needs a phonon too, so it is slow and non-radiative trap recombination wins.

Q. What does a sloping quasi-Fermi level mean?
A. Net current of that carrier type. A flat quasi-Fermi level means no current, even if the carrier concentration is far from equilibrium.

1.10 Poisson's equation and the continuity equation

Two equations govern every device in this course. Learn what each one "says" in words; the maths follows.

Poisson's equation: charge creates field

POISSON'S EQUATION (1-D)

$$\frac{dE}{dx} = \frac{\rho(x)}{\varepsilon_s} = -\frac{d^2\varphi}{dx^2} \qquad \rho = q(p - n + N_d^+ - N_a^-)$$

In words: wherever there is net charge, the field changes. In a neutral region $\rho = 0$, so the field is constant. In a depletion region (only fixed ions) ρ is constant, so the field changes linearly and the potential parabolically. That one sentence gives you the field and potential plots of the PN junction in Module 2.

Continuity equation: carriers are conserved

Take a thin slice of width dx. The hole count in it can change for only three reasons: more holes flow in than out, holes are generated, or holes recombine.

Net inflow rate per unit volume = $[J_p(x) - J_p(x + dx)]/(q\,dx) = -(1/q)(\partial J_p/\partial x)$.
So $\partial p/\partial t = -(1/q)(\partial J_p/\partial x) + G_p - p/\tau_{pt}$, where the last term is the total recombination rate.
Insert $J_p = qp\mu_p E - qD_p \partial p/\partial x$ and use $R - G_{thermal} = \delta p/\tau_p$ (low-level injection).

CONTINUITY EQUATIONS

$$\partial p/\partial t = -\frac{1}{q}\,\partial J_p/\partial x + G_p - R_p \qquad \partial n/\partial t = +\frac{1}{q}\,\partial J_n/\partial x + G_n - R_n$$

Minority holes in n-type: $\partial \delta p/\partial t = D_p\,\partial^2 \delta p/\partial x^2 - \mu_p E\,\partial \delta p/\partial x + g' - \delta p/\tau_p$

(diffusion − drift + external generation − recombination; E assumed uniform)

KEY IDEA

The sign difference between the hole and electron equations is only because electrons carry −q. Everything else is "what flows in, what is made, what is lost". Most exam questions use special cases: uniform (no $\partial/\partial x$ terms), steady state ($\partial/\partial t = 0$), or zero field.

1.11 Steady-state solution: the diffusion length

The classic problem: light (or a junction) keeps injecting excess holes at the surface $x = 0$ of a long n-type bar. No field, no generation inside the bar ($x > 0$), steady state. How far do the holes get before they recombine?

Continuity with $\partial/\partial t = 0$, $E = 0$, $g' = 0$: $D_p \, d^2\delta p/dx^2 - \delta p/\tau_p = 0$.
Define $L_p = \sqrt{(D_p \tau_p)}$: $d^2\delta p/dx^2 = \delta p/L_p^2$.
General solution: $\delta p(x) = A\, e^{-x/L_p} + B\, e^{+x/L_p}$.
Long bar: δp must stay finite as $x \to \infty \Rightarrow B = 0$. At $x = 0$, $\delta p = \delta p(0) \Rightarrow A = \delta p(0)$.

DIFFUSION LENGTH

$$\delta p(x) = \delta p(0)\, e^{-x/L_p} \qquad L_p = \sqrt{(D_p \tau_p)} \qquad J_p(x) = \frac{qD_p \delta p(0)}{L_p}\, e^{-x/L_p}$$

L_p is the average distance an excess minority carrier diffuses before recombining. If the bar is much shorter than L_p (a "short" region, $W \ll L_p$), recombination is negligible, the profile becomes a straight line, and the current is $qD_p\delta p(0)/W$. Keep both cases in mind: the long case gives the ideal diode, the short case gives the BJT base.

SOLVED EXAMPLE 1.9 · DIFFUSION LENGTH KTU FAVOURITE

n-type Si with $\mu_p = 480$ cm²/V·s and $\tau_p = 1$ µs. Steady injection at $x = 0$ holds $\delta p(0) = 10^{14}$ cm^{-3}. Find L_p, the hole diffusion current density at $x = 0$, and δp at $x = 50$ µm.

- $D_p = V_T \mu_p = 12.43$ cm²/s. $L_p = \sqrt{(12.43 \times 10^{-6})} = 3.53 \times 10^{-3}$ cm = 35.3 µm.
- $J_p(0) = qD_p\delta p(0)/L_p = 1.602 \times 10^{-19} \times 12.43 \times 10^{14}/3.53 \times 10^{-3} = 0.0565$ A/cm².
- $\delta p(50\ \mu m) = 10^{14} \times \exp(-50/35.3) = 2.42 \times 10^{13}$ cm^{-3}.

$L_p = 35.3$ µm, $J_p(0) = 56.5$ mA/cm², $\delta p(50\ \mu m) = 2.42 \times 10^{13}$ cm^{-3}

1.12 Quasi-Fermi levels

In equilibrium one Fermi level describes both carriers: $n_0 = n_i e^{(E_F - E_i)/kT}$ and $p_0 = n_i e^{(E_i - E_F)/kT}$. With excess carriers present, $np > n_i^2$ and no single E_F can satisfy both. The fix: give each carrier its own "Fermi level", defined so the familiar formulas still hold.

QUASI-FERMI LEVELS

$$n = n_0 + \delta n = n_i\, e^{(E_{Fn} - E_i)/kT} \qquad p = p_0 + \delta p = n_i\, e^{(E_i - E_{Fp})/kT}$$

$$np = n_i^2\, e^{(E_{Fn} - E_{Fp})/kT}$$

The **separation $E_{Fn} - E_{Fp}$** measures how far the system is from equilibrium. It is zero in equilibrium. In a forward-biased diode it equals qV_a, which is the cleanest way to explain the e^{V/V_T} law. A gradient in a quasi-Fermi level means current: $J_n = n\mu_n\, dE_{Fn}/dx$.

SOLVED EXAMPLE 1.10 · QUASI-FERMI LEVELS

Use the sample of Example 1.8 ($N_d = 10^{16}$ cm^{-3}, $\delta n = \delta p = 10^{14}$ cm^{-3}). Locate E_{Fn} and E_{Fp} relative to E_i and compare with the dark E_F.

- Dark: $E_F - E_i = 0.0259 \ln(10^{16}/1.5 \times 10^{10}) = 0.3473$ eV.
- $E_{Fn} - E_i = 0.0259 \ln(1.01 \times 10^{16}/1.5 \times 10^{10}) = 0.3476$ eV, almost unchanged (majority carriers barely changed).
- $p \approx 10^{14}$: $E_i - E_{Fp} = 0.0259 \ln(10^{14}/1.5 \times 10^{10}) = 0.228$ eV, so E_{Fp} sits 0.228 eV *below* E_i. It moved by more than half an eV.

$E_{Fn} = E_i + 0.348$ eV, $E_{Fp} = E_i - 0.228$ eV; splitting ≈ 0.576 eV

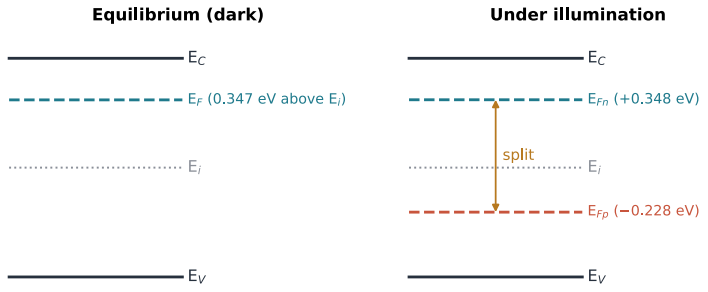


Fig 1.10 Example 1.10 drawn to scale: light barely moves E_{Fn} but drags E_{Fp} far down.

✓ Module 1: Quick revision

Bands: E_g(Si) = 1.12 eV, indirect; GaAs 1.42 eV, direct. **Fermi–Dirac:** $f = 1/(1 + e^{(E-E_F)/kT})$; Boltzmann if $|E - E_F| > 3kT$.

Concentrations: $n_0 = N_c e^{-(E_C-E_F)/kT} = n_i e^{(E_F-E_i)/kT}$; **mass action** $n_0 p_0 = n_i^2$ (doping-independent). **DOS:** $g_c \propto (m_n^*)^{3/2}\sqrt{(E - E_C)}$. $N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$.

Neutrality: $n_0 + N_a = p_0 + N_d \Rightarrow n_0 = (N_d - N_a)/2 + \sqrt{[((N_d - N_a)/2)^2 + n_i^2]}$.

E_F shift: $kT \ln(N/n_i)$, 59.6 meV/decade; high T ⇒ $E_F \to E_i$. E_i = mid-gap + ¾kT $\ln(m_p^*/m_n^*)$.

Hall: $V_H = IB/(qpd)$; $R_H = \pm 1/(qp \text{ or } qn)$; $\mu = \sigma|R_H|$. Sign gives type. **Drift:** $J = q(n\mu_n + p\mu_p)E$; $\mu = q\tau_c/m^*$; $v_{sat} \approx 10^7$ cm/s.

Diffusion: $J_n = +qD_n dn/dx$, $J_p = -qD_p dp/dx$. Einstein $D/\mu = kT/q$.

Lifetime: $R = \delta p/\tau_p$; steady $\delta p = G\tau_p$; decay $e^{-t/\tau}$. **Diffusion length:** $L = \sqrt{(D\tau)}$; $\delta p(x) = \delta p(0)e^{-x/L}$; $J(0) = qD\delta p(0)/L$.

Poisson: $dE/dx = \rho/\varepsilon_s$. **Continuity:** $\partial p/\partial t = -(1/q)\partial J_p/\partial x + G - R$.

Quasi-Fermi: $np = n_i^2 e^{(E_{Fn}-E_{Fp})/kT}$; split = qV_a in a diode.

Practice: short questions (3 marks)

1. Distinguish between direct and indirect bandgap semiconductors with E-k sketches.
2. State the mass-action law. Why is it independent of doping?
3. Find the probability that a state 0.2 eV below E_F is empty at 300 K. [4.3×10^{-4}]
4. Si is doped with 10^{17} cm^{-3} boron. Find n_0, p_0 and $E_i - E_F$. [2.25×10^3, 10^{17}, 0.407 eV]
5. Explain how the Hall effect identifies the carrier type.
6. Derive the Einstein relation.
7. Define minority carrier lifetime and diffusion length. How are they related?
8. What are quasi-Fermi levels? Why are they needed?

Practice: long questions (9/14 marks)

1. Derive n_0 from g(E) and f(E); hence prove $n_0 p_0 = n_i^2$.
2. Explain n and E_F vs temperature for n-type Si (freeze-out, extrinsic, intrinsic regions).
3. Derive V_H and show how n and μ are found. With d = 10 μm, I = 1 mA, B = 0.2 T, V_H = −2 mV: find type and concentration. [n-type, 6.24×10^{16} cm^{-3}]
4. Derive the hole continuity equation and solve its steady-state form for a long sample with injection at x = 0.
5. n-type Si with τ_p = 0.5 μs, D_p = 10 cm²/s has $\delta p(0) = 10^{13}$ cm^{-3} held at x = 0. Find L_p and $J_p(0)$. [22.4 μm, 7.17 mA/cm²]
6. Explain drift and diffusion currents, scattering mechanisms and velocity saturation.

02

PN Junction & Bipolar Junction Transistor

> *The PN junction is inside every diode, BJT, solar cell, LED and MOSFET (source and drain are junctions too). "Derive the diode equation" and "why is β large?" are classic viva and interview questions.*

SYLLABUS CHECKLIST

- ☐ PN junction in equilibrium: built-in potential, depletion width, field and potential profiles
- ☐ Forward and reverse bias
- ☐ Derivation of the ideal diode equation
- ☐ Junction (depletion) capacitance and diffusion capacitance
- ☐ Breakdown basics: Zener and avalanche
- ☐ BJT: structure, current components
- ☐ Emitter injection efficiency, base transport factor, α and β
- ☐ Minority carrier distributions; Early effect

12 hours · CO2 · Analyse PN junction and BJT operation

2.1 The PN junction in equilibrium

Imagine joining a p-type piece (lots of holes) to an n-type piece (lots of electrons). The moment they touch, holes diffuse into the n side and electrons diffuse into the p side, just like two gases mixing. But each carrier that leaves exposes a **fixed ion** it can't take along: negative acceptor ions on the p side, positive donor ions on the n side. These ions form a thin **space-charge (depletion) region**, empty of mobile carriers, and their charge sets up an electric field pointing from n to p.

That field pushes carriers back (drift). Very quickly, drift exactly balances diffusion for each carrier type, and the net current is zero. The potential difference that builds up across the junction is the **built-in potential** V_{bi}.

> **THINK OF IT LIKE**
>
> Two crowded rooms with a door between them. People drift across, but every person who crosses leaves behind a "coat" (the fixed ion) near the door, and the pile of coats starts pushing newcomers back. Soon crossings each way balance. The pile of coats is the depletion region; the push is the built-in field.

Why a potential, and how big? In equilibrium E_F must be flat through the whole structure. On the p side E_F sits near E_V; on the n side near E_C. The only way to line them up is for the bands to bend across the junction by exactly the difference in Fermi levels of the two isolated pieces:

p side: $E_i - E_F = kT \ln(N_a/n_i)$. n side: $E_F - E_i = kT \ln(N_d/n_i)$.
qV_{bi} = sum of the two = $kT \ln(N_a N_d/n_i^2)$.

BUILT-IN POTENTIAL

$$V_{bi} = \frac{kT}{q} \ln \frac{N_a N_d}{n_i^2}$$

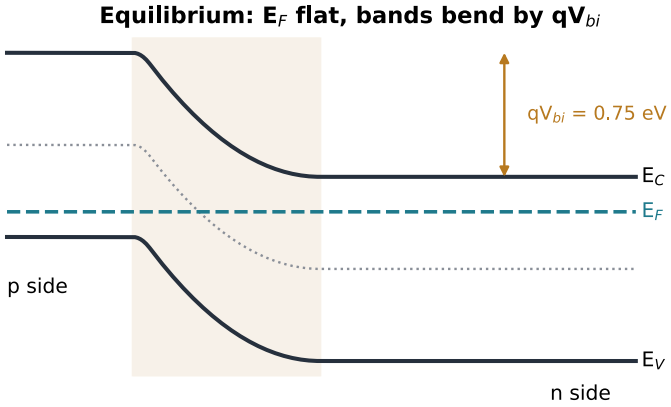


Fig 2.1 Equilibrium band diagram for $N_a = 10^{17}$, $N_d = 10^{16}$ cm^{-3} (drawn to scale). The shaded strip is the depletion region.

COMMON MISTAKE

"Connect a voltmeter across a diode and it should read V_{bi}." It reads zero. The metal contacts form their own contact potentials that exactly cancel V_{bi}; otherwise you could draw power from a diode sitting on a table.

SOLVED EXAMPLE 2.1 · BUILT-IN POTENTIAL

A Si junction has $N_a = 10^{17}$ cm^{-3} and $N_d = 10^{16}$ cm^{-3} at 300 K. Find V_{bi}.

- $N_aN_d/n_i^2 = 10^{33}/2.25 \times 10^{20} = 4.44 \times 10^{12}$; ln = 29.12.
- $V_{bi} = 0.0259 \times 29.12 = 0.754$ V.

V_{bi} = 0.754 V (this junction is used in every example of this module)

2.2 Depletion width, field and potential profiles

We now solve Poisson's equation using the **depletion approximation**: inside $-x_p < x < x_n$ there are no mobile carriers at all (only ions); outside it the material is perfectly neutral. The charge density is then two rectangles: $-qN_a$ on the p side and $+qN_d$ on the n side.

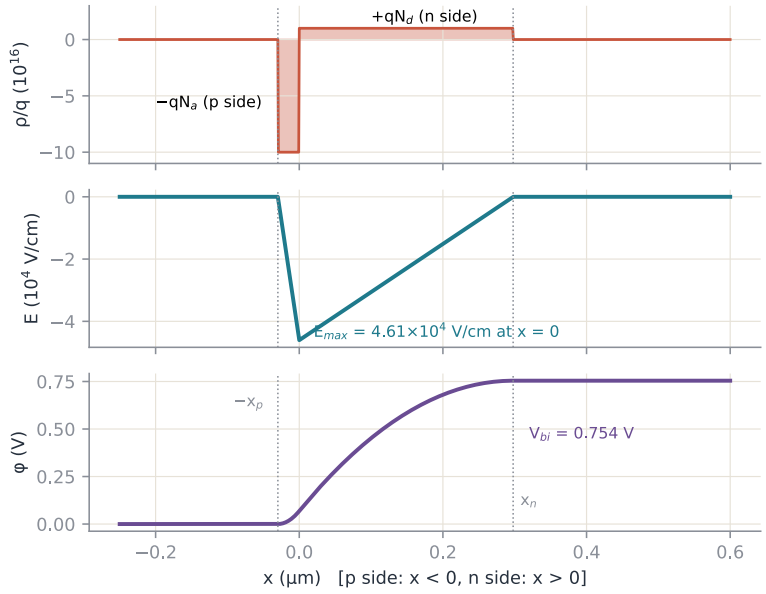


Fig 2.2 Charge, field and potential for the example junction (computed). Rectangles of charge give a triangular field and a parabolic potential. The lighter n side carries most of the width.

Derivation

p side ($-x_p < x < 0$): $dE/dx = -qN_a/\varepsilon_s$, with $E(-x_p) = 0 \Rightarrow E(x) = -(qN_a/\varepsilon_s)(x + x_p)$.
n side ($0 < x < x_n$): $dE/dx = +qN_d/\varepsilon_s$, with $E(x_n) = 0 \Rightarrow E(x) = -(qN_d/\varepsilon_s)(x_n - x)$.
E must be continuous at $x = 0 \Rightarrow N_ax_p = N_dx_n$: **total charge on each side is equal**, so the more lightly doped side depletes deeper.

Peak field at x = 0: $|E_{max}| = qN_dx_n/\varepsilon_s = qN_ax_p/\varepsilon_s$.
V_{bi} = area of the field triangle = ½$|E_{max}|$W, with $W = x_n + x_p$. Eliminating x_n, x_p gives the width.

DEPLETION REGION (STEP JUNCTION)

$$W = \sqrt{\left[\frac{2\varepsilon_s(V_{bi} + V_R)}{q} \cdot \frac{N_a + N_d}{N_aN_d}\right]}$$

$$x_n = W\frac{N_a}{N_a + N_d} \qquad x_p = W\frac{N_d}{N_a + N_d} \qquad |E_{max}| = \frac{2(V_{bi} + V_R)}{W}$$

One-sided p^+n junction ($N_a \gg N_d$): $W \approx \sqrt{[2\varepsilon_s(V_{bi} + V_R)/(qN_d)]}$, almost all on the n side. Forward bias: replace V_R by $-V_a$.

SOLVED EXAMPLE 2.2 · DEPLETION WIDTH AND FIELD KTU FAVOURITE

For the junction of Example 2.1, find W, x_n, x_p and E_{max} at zero bias and at V_R = 5 V.

- $(N_a + N_d)/(N_aN_d) = 1.1 \times 10^{17}/10^{33} = 1.1 \times 10^{-16}$ cm³.
- V = 0: $W = \sqrt{(2 \times 1.036 \times 10^{-12} \times 0.754 \times 1.1 \times 10^{-16}/1.602 \times 10^{-19})} = 3.28 \times 10^{-5}$ cm = 0.328 µm.
- $x_n = W \times 10/11 = 0.298$ µm; $x_p = W/11 = 0.0298$ µm. $E_{max} = 2 \times 0.754/3.28 \times 10^{-5} = 4.61 \times 10^4$ V/cm.
- V_R = 5 V: W scales by $\sqrt{(5.754/0.754)} = 2.76 \Rightarrow W = 0.905$ µm, $x_n = 0.823$ µm, $x_p = 0.082$ µm, $E_{max} = 2 \times 5.754/9.05 \times 10^{-5} = 1.27 \times 10^5$ V/cm.

0 V: W = 0.328 µm, $E_{max} = 4.61 \times 10^4$ V/cm · 5 V: W = 0.905 µm, $E_{max} = 1.27 \times 10^5$ V/cm

2.3 Forward and reverse bias

An applied voltage drops almost entirely across the depletion region (the neutral regions are good conductors in comparison). So bias simply **changes the barrier height** from qV_{bi} to $q(V_{bi} - V_a)$.

- **Forward bias** (p positive): barrier lowered, depletion narrows. The diffusion flow, which depends exponentially on barrier height, grows hugely while the drift flow stays the same. Large net current. Electrons are **injected** into the p side and holes into the n side, where they are minority carriers.
- **Reverse bias** (n positive): barrier raised, depletion widens, field grows. Diffusion is choked off. Only the tiny drift of minority carriers that wander to the edge and get swept across remains: the **reverse saturation current** I_s.

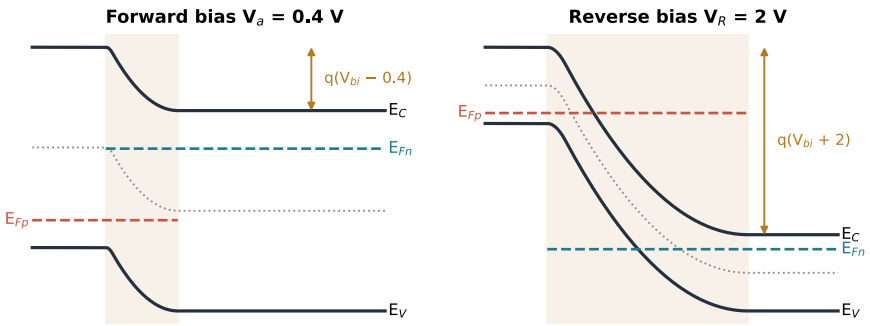


Fig 2.3 Under bias the Fermi level splits into quasi-Fermi levels separated by qV_a. Forward: barrier lowered. Reverse: barrier raised, depletion wider.

2.4 Derivation of the ideal diode equation

Assumptions (state them in the exam, they carry marks): (1) abrupt junction with depletion approximation; (2) Boltzmann statistics; (3) **low-level injection**; (4) no generation or recombination inside the depletion region; (5) long neutral regions (length ≫ diffusion length); (6) steady state.

Step 1: boundary condition (law of the junction)

In equilibrium, $n_{n0} = n_{p0}e^{V_{bi}/V_T}$ (the electron populations on the two sides differ by the Boltzmann factor of the barrier). Under bias the barrier is $V_{bi} - V_a$ and the majority density hardly changes (low-level injection), so the minority density at the depletion edge rises exponentially:

LAW OF THE JUNCTION

$$p_n(x_n) = p_{n0}\, e^{V_a/V_T} \qquad n_p(-x_p) = n_{p0}\, e^{V_a/V_T}$$

Step 2: minority carrier profiles

In the neutral n region there is no field and no generation, so the excess holes obey the steady-state diffusion equation of section 1.11: $d^2\delta p/dx^2 = \delta p/L_p^2$. With $\delta p \to 0$ deep inside and $\delta p(x_n) = p_{n0}(e^{V_a/V_T} - 1)$:

$$\delta p_n(x) = p_{n0}(e^{V_a/V_T} - 1)\, e^{-(x - x_n)/L_p} \quad (x \geq x_n)$$

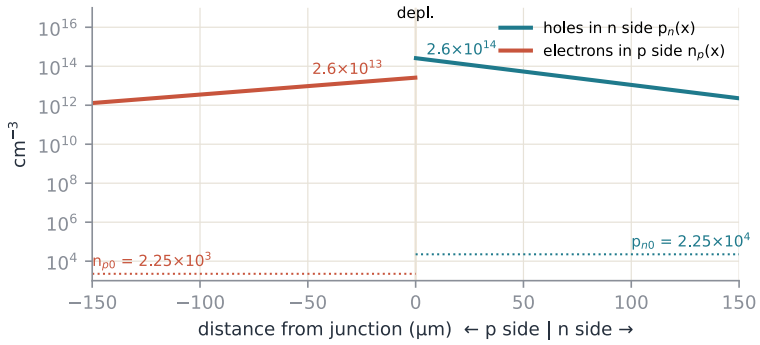


Fig 2.4 Minority carriers at V_a = 0.6 V (log scale, computed with L_n = 50 µm, L_p = 31.6 µm). Injected carriers decay over a diffusion length.

Step 3: currents and total

Hole diffusion current at x_n: $J_p = -qD_p\, d\delta p/dx\,|_{x_n} = (qD_p p_{n0}/L_p)(e^{V_a/V_T} - 1)$.
Similarly, electron current at $-x_p$: $J_n = (qD_n n_{p0}/L_n)(e^{V_a/V_T} - 1)$.
No recombination in the depletion region ⇒ each current is the same at both edges, so total $J = J_p(x_n) + J_n(-x_p)$. Use $p_{n0} = n_i^2/N_d$, $n_{p0} = n_i^2/N_a$.

IDEAL (SHOCKLEY) DIODE EQUATION

$$I = I_s(e^{V_a/V_T} - 1) \qquad I_s = Aqn_i^2\left(\frac{D_n}{L_n N_a} + \frac{D_p}{L_p N_d}\right)$$

Short diode (neutral length $W \ll L$): replace L by W. Real diodes: $I = I_s(e^{V/\eta V_T} - 1)$, ideality η = 1 to 2 ($\eta \to 2$ when depletion-region recombination dominates).

KEY IDEA

The diode current is set by **minority carriers diffusing in the neutral regions**, and $I_s \propto n_i^2/N$. So the **lightly doped side** dominates the current (more of its minority carriers), and I_s is extremely temperature sensitive (n_i^2).

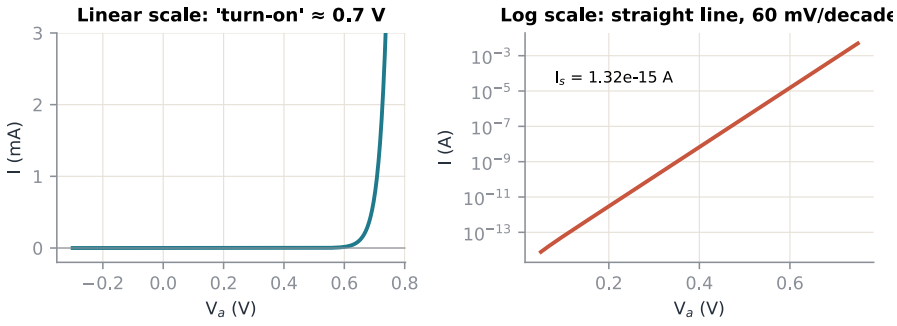


Fig 2.5 Computed I-V of the example diode (A = 10^{-4} cm²). On a log scale the forward current is a straight line rising one decade per 59.6 mV.

SOLVED EXAMPLE 2.3 · IDEAL DIODE CURRENT KTU FAVOURITE

The junction of Example 2.1 has A = 10^{-4} cm², D_n = 25 cm²/s, D_p = 10 cm²/s, $\tau_n = \tau_p$ = 1 µs (long diode). Find I_s, the current at 0.6 V and 0.7 V, the fraction carried by holes, and the voltage for 1 mA.

- $L_n = \sqrt{(25 \times 10^{-6})} = 50$ µm; $L_p = \sqrt{(10 \times 10^{-6})} = 31.6$ µm.
- $D_n/(L_n N_a) = 25/(5 \times 10^{-3} \times 10^{17}) = 5 \times 10^{-14}$; $D_p/(L_p N_d) = 10/(3.16 \times 10^{-3} \times 10^{16}) = 3.16 \times 10^{-13}$ (cm⁴/s).
- $J_s = 1.602 \times 10^{-19} \times 2.25 \times 10^{20} \times 3.66 \times 10^{-13} = 1.32 \times 10^{-11}$ A/cm², so $I_s = 1.32 \times 10^{-15}$ A.
- $I(0.6) = 1.32 \times 10^{-15} \times e^{23.17} = 15.2$ µA. I(0.7) = 0.722 mA (×47.5 for +0.1 V).
- Hole fraction = 3.16/(3.16 + 0.5) = 86 %: the lightly doped n side dominates. V(1 mA) = 0.0259 ln($10^{-3}/1.32 \times 10^{-15}$) = 0.708 V.

I_s = 1.32 fA; 15.2 µA at 0.6 V; 0.722 mA at 0.7 V; 86 % hole current; 0.708 V for 1 mA

SOLVED EXAMPLE 2.4 · CHECK LOW-LEVEL INJECTION

For the same diode at V_a = 0.6 V, find the minority concentrations at both depletion edges. Is low-level injection valid?

- $p_{n0} = 2.25 \times 10^4$; $p_n(x_n) = 2.25 \times 10^4 \times e^{0.6/0.0259} = 2.59 \times 10^{14}$ cm^{-3}.
- $n_{p0} = 2.25 \times 10^3$; $n_p(-x_p) = 2.59 \times 10^{13}$ cm^{-3}.
- $2.59 \times 10^{14} \ll N_d = 10^{16}$ and $2.59 \times 10^{13} \ll N_a = 10^{17} \Rightarrow$ valid (just). Near 0.7 V the n side approaches high-level injection.

2.59×10^{14} and 2.59×10^{13} cm^{-3}; low-level injection holds

SOLVED EXAMPLE 2.5 · THE 60 MV RULE INTERVIEW

A diode carries 1 mA at 0.65 V. Find the current at 0.70 V, and the voltage change needed for a 10× change in current (η = 1, 300 K).

- $I_2/I_1 = e^{0.05/0.0259} = 6.89 \Rightarrow I = 6.89$ mA.
- ΔV for 10× = $V_T \ln 10 = 59.6$ mV.

6.89 mA; 59.6 mV per decade

2.5 Junction and diffusion capacitance

Junction (depletion) capacitance

The depletion region is two sheets of charge separated by a gap, which is a capacitor. Increase V_R and W widens, uncovering more ion charge dQ for a change dV. Because the charge sits at the two edges of W, the capacitance is that of a parallel-plate capacitor with spacing W:

JUNCTION CAPACITANCE

$$C_j = \frac{\varepsilon_s A}{W} = A\sqrt{\left[\frac{q\varepsilon_s N_a N_d}{2(V_{bi}+V_R)(N_a+N_d)}\right]} \propto (V_{bi}+V_R)^{-1/2}$$

$$\text{One-sided: } \frac{1}{(C_j/A)^2} = \frac{2(V_{bi}+V_R)}{q\varepsilon_s N_d} \Rightarrow \text{slope gives } N_d\text{, intercept gives } -V_{bi}$$

This voltage-controlled capacitance is exploited in **varactor diodes** for VCOs and tuners. The $1/C^2$ plot is the standard lab method to measure doping (C-V profiling).

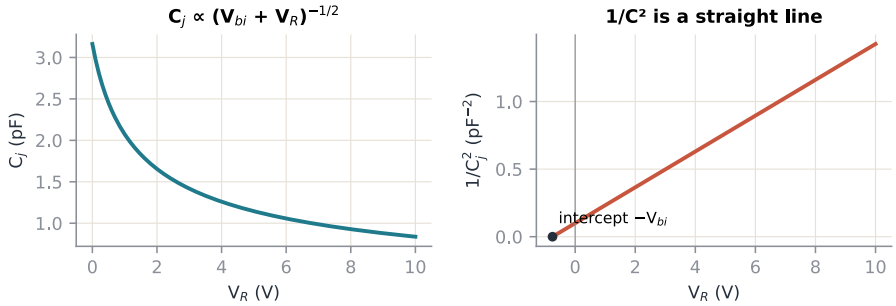


Fig 2.6 C_j of the example diode (A = 10^{-4} cm²) and its $1/C^2$ straight line, whose intercept is $-V_{bi}$ = −0.754 V.

Diffusion (storage) capacitance

Under forward bias a large charge of injected minority carriers is **stored** in the neutral regions (the areas under the curves in Fig 2.4). Raise V_a a little and this stored charge grows: another capacitance. For a p^+n diode the stored hole charge is $Q_p = I_p\tau_p$, and the small-signal analysis gives:

DIFFUSION CAPACITANCE AND CONDUCTANCE (LONG P⁺N DIODE)

$$g_d = \frac{I_D}{V_T} \qquad C_d = \frac{\tau_p I_D}{2V_T}$$

(the ½ appears for a long diode; a short diode has $C_d = \tau_T I_D/V_T$, τ_T = transit time)

	Junction capacitance C_j	Diffusion capacitance C_d
Origin	Ion charge in depletion region	Stored minority carriers in neutral regions
Dominant in	Reverse bias (and small forward bias)	Forward bias
Depends on	$(V_{bi} + V_R)^{-1/2}$	$\propto I_D$, $\propto$ lifetime
Typical value	pF	nF at mA currents

SOLVED EXAMPLE 2.6 · BOTH CAPACITANCES

(a) For the example diode ($A = 10^{-4}$ cm²) find C_j at 0 V and at $V_R = 5$ V. (b) A p^+n diode with $\tau_p = 1$ µs carries 1 mA forward. Find g_d and C_d.

▸ (a) $C_j/A = \varepsilon_s/W = 1.036 \times 10^{-12}/3.28 \times 10^{-5} = 3.16 \times 10^{-8}$ F/cm² ⇒ $C_j = 3.16$ pF. At 5 V, W = 0.905 µm ⇒ $C_j = 1.14$ pF.

▸ (b) $g_d = 1$ mA/25.9 mV = 38.6 mS ($r_d = 25.9$ Ω). $C_d = 10^{-6} \times 10^{-3}/(2 \times 0.0259) = 19.3$ nF.

C_j = 3.16 pF (0 V), 1.14 pF (5 V); g_d = 38.6 mS, C_d = 19.3 nF

EXAM TIP

Stored charge is why a diode cannot switch off instantly: the stored minority carriers must be removed first (reverse recovery time $\propto \tau$). Fast-recovery diodes use gold doping to cut τ; Schottky diodes (Module 3) have almost no stored charge.

2.6 Junction breakdown: Zener and avalanche

Push reverse bias far enough and the current suddenly shoots up. The junction is not destroyed (unless it overheats); the field has simply become strong enough for one of two mechanisms:

- **Zener (tunnelling) breakdown**: in **heavily doped** junctions the depletion region is very thin (≈ 10 nm). At a few volts the valence band on the p side lines up opposite empty conduction-band states on the n side, only a thin barrier apart, and electrons **tunnel** straight through. Typically below about 5 V. Temperature coefficient **negative** (E_g shrinks with T, so tunnelling gets easier).
- **Avalanche breakdown**: in **lightly doped**, wider junctions, a carrier accelerated by the field gains enough energy to knock an electron out of a bond (**impact ionisation**), creating a new electron-hole pair. These carriers ionise more, and the multiplication runs away. Typically above about 6–8 V. Temperature coefficient **positive** (more lattice vibration means carriers lose energy more often, so more voltage is needed).

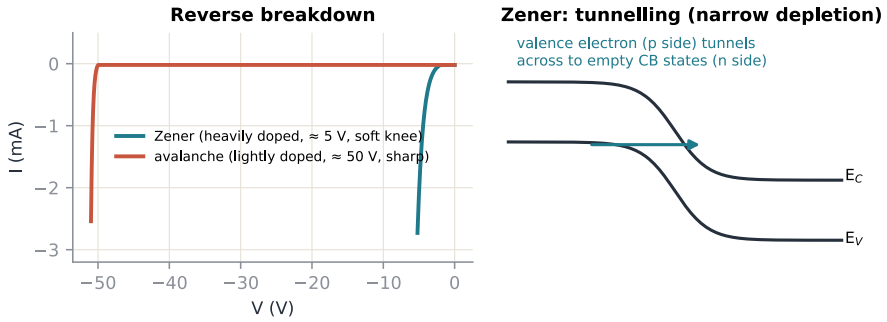


Fig 2.7 Left: qualitative reverse characteristics. Right: the band picture of Zener tunnelling.

AVALANCHE BREAKDOWN VOLTAGE (ONE-SIDED P⁺N)

$$\text{Breakdown when } E_{max} = E_{crit}\text{:}\quad V_B \approx \frac{\varepsilon_s E_{crit}^{\,2}}{2qN_d} \quad (V_B \gg V_{bi})$$

SOLVED EXAMPLE 2.7 · BREAKDOWN VOLTAGE

A p^+n Si junction has $N_d = 10^{16}$ cm^{-3} and $E_{crit} = 4 \times 10^5$ V/cm. Estimate V_B. What happens if N_d is raised to 10^{17}?

▸ $V_B = 1.036 \times 10^{-12} \times (4 \times 10^5)^2/(2 \times 1.602 \times 10^{-19} \times 10^{16}) = 51.7$ V.

▸ $V_B \propto 1/N_d \Rightarrow$ about 5.2 V at 10^{17} (in reality E_{crit} rises with doping, so it is somewhat higher, and tunnelling begins to take over).

$V_B \approx 51.7$ V; roughly 10× lower with 10× more doping

INTERVIEW CORNER

Q. A 5.6 V Zener diode is popular for references. Why that value?

A. Near 5–6 V both mechanisms contribute, and their opposite temperature coefficients roughly cancel, giving a near-zero tempco.

Q. Why does reverse current saturate?

A. It is the drift of thermally generated minority carriers reaching the junction; their supply is fixed by n_i^2/N and diffusion length, not by the voltage.

Q. Which side of a p^+n junction does the depletion region extend into?

A. Almost entirely the lightly doped n side, because $N_a x_p = N_d x_n$.

2.7 BJT structure and current components

A bipolar junction transistor is two PN junctions back to back, sharing a **very thin** middle region. In an npn: a heavily doped n^+ **emitter**, a thin, moderately doped p **base**, and a lightly doped n **collector** (larger, to handle voltage and heat).

The trick. Forward-bias the E-B junction: the emitter injects a flood of electrons into the base. The base is so thin ($\ll L_n$) that almost none of them recombine; they diffuse across and reach the reverse-biased C-B junction, whose field sweeps them into the collector. A small base current (to replace the few that recombine and to supply holes injected back into the emitter) controls a large collector current. That is gain.

THINK OF IT LIKE

A waterfall. The emitter is a reservoir pushing water over a low wall (forward-biased E-B). The base is a very short ledge. The collector junction is the cliff edge: whatever reaches it falls, no matter how high the cliff (V_{CB}). The base current is the small amount of water lost on the ledge.

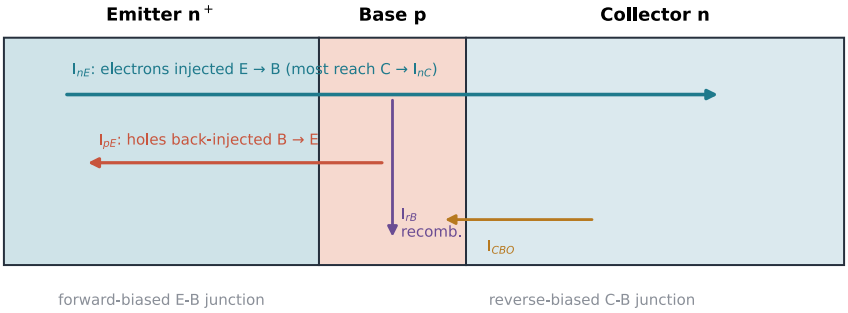


Fig 2.8 Current components in an npn BJT in forward-active mode (electron flows shown; conventional currents are opposite for electrons).

TERMINAL CURRENTS

$$I_E = I_{nE} + I_{pE} \qquad I_C = I_{nC} + I_{CBO} \qquad I_B = I_E - I_C = I_{pE} + I_{rB} - I_{CBO}$$

Mode	E-B junction	C-B junction	Use
Forward active	Forward	Reverse	Amplifier
Saturation	Forward	Forward	Switch ON
Cut-off	Reverse	Reverse	Switch OFF
Reverse active	Reverse	Forward	Rarely (poor gain)

2.8 Injection efficiency, transport factor, α and β

The common-base current gain $\alpha = I_C/I_E$ (ignoring I_{CBO}) is a chain of "efficiencies", each the fraction that survives one stage:

GAIN FACTORS

$$\gamma = \frac{I_{nE}}{I_{nE} + I_{pE}} \quad \text{(emitter injection efficiency)} \qquad \alpha_T = \frac{I_{nC}}{I_{nE}} \quad \text{(base transport factor)}$$

$$\alpha = \gamma\, \alpha_T\, \delta \quad (\delta = \text{recombination factor of the E-B depletion region}, \approx 1) \qquad \beta = \frac{I_C}{I_B} = \frac{\alpha}{1-\alpha}, \quad \alpha = \frac{\beta}{\beta+1}$$

Using the diode-current expressions of section 2.4 for each side (short base of width W_B, long emitter):

PHYSICAL EXPRESSIONS

$$\gamma \approx \frac{1}{1 + \frac{N_B D_E W_B}{N_E D_B L_E}} \qquad \alpha_T = \operatorname{sech}(W_B/L_B) \approx 1 - \frac{W_B^2}{2L_B^2}$$

(D_E: hole diffusivity in emitter, D_B: electron diffusivity in base, L_E, L_B: their diffusion lengths. For a short emitter replace L_E by W_E.)

KEY IDEA

To get high β: make $\gamma \to 1$ by doping the **emitter much more heavily than the base** ($N_E \gg N_B$), and make $\alpha_T \to 1$ with a **thin base** ($W_B \ll L_B$). Because $\beta = \alpha/(1-\alpha)$, a tiny change in α makes a big change in β.

SOLVED EXAMPLE 2.8 · B FROM PHYSICS GATE

An npn BJT has $N_E = 10^{18}$, $N_B = 10^{17}$ cm^{-3}, $D_E = 2$, $D_B = 20$ cm²/s, $L_E = 1$ μm, $L_B = 10$ μm, $W_B = 0.5$ μm. Find γ, α_T, α and β (δ = 1). Repeat for $W_B = 1$ μm.

- $N_B D_E W_B/(N_E D_B L_E) = (10^{17} \times 2 \times 0.5)/(10^{18} \times 20 \times 1) = 0.005 \Rightarrow \gamma = 1/1.005 = 0.99502$.
- $\alpha_T = 1 - 0.5^2/(2 \times 10^2) = 0.99875$.
- $\alpha = 0.99502 \times 0.99875 = 0.99378 \Rightarrow \beta = 0.99378/0.00622 = 160$.
- $W_B = 1$ μm: ratio 0.01 ⇒ $\gamma = 0.99010$; $\alpha_T = 0.995$; $\alpha = 0.98515 \Rightarrow \beta = 66$.

$\gamma = 0.995$, $\alpha_T = 0.9988$, $\alpha = 0.9938$, $\beta \approx 160$; doubling W_B drops β to ≈ 66

SOLVED EXAMPLE 2.9 · A–B CONVERSION

A BJT has β = 100 and $I_E = 1$ mA. Find α, I_C and I_B. What α is needed for β = 200?

- $\alpha = 100/101 = 0.990$. $I_C = \alpha I_E = 0.990$ mA; $I_B = I_E/(\beta + 1) = 9.90$ μA.
- $\beta = 200 \Rightarrow \alpha = 200/201 = 0.995$: only half a percent more.

$\alpha = 0.990$, $I_C = 0.990$ mA, $I_B = 9.90$ μA; $\alpha = 0.995$ for $\beta = 200$

INTERVIEW CORNER

Q. Why can't you make a BJT by soldering two diodes back to back?

A. The base would be far wider than the diffusion length, so every injected carrier recombines before reaching the collector: $\alpha_T \approx 0$, no transistor action.

Q. Why is the emitter doped more heavily than the collector? Can you swap E and C?

A. Heavy emitter doping gives $\gamma \approx 1$. The collector is light for high breakdown voltage and low C_{CB}. Swapped (reverse active), β drops to around 1–5.

Q. What is the physical origin of the Early effect?

A. Base-width modulation: the C-B depletion region widens with V_{CB}, shrinking the neutral base and steepening the minority carrier gradient.

Q. Why is I_C almost independent of V_{CB}?

A. The collector junction only collects; I_C is set by how many carriers the emitter injects (V_{BE}). Any V_{CB} that keeps the junction reverse-biased sweeps them all.

2.9 Minority carrier distributions in a BJT

Apply the law of the junction at each edge of the base and solve the diffusion equation in each region, exactly as for the diode:

- **Base** (electrons): at the E-B edge $n_B(0) = n_{B0}e^{V_{BE}/V_T}$; at the reverse-biased C-B edge $n_B(W_B) = n_{B0}e^{-V_{CB}/V_T} \approx 0$. General solution: $n_B(x) - n_{B0} = A \sinh[(W_B - x)/L_B] + B \sinh(x/L_B)$. For $W_B \ll L_B$ it is almost a **straight line**, so the collector current is a pure diffusion current $I_C \approx qAD_B n_{B0}e^{V_{BE}/V_T}/W_B$.
- **Emitter** (holes): decay from $p_{E0}e^{V_{BE}/V_T}$ over L_E, like a diode. Small because N_E is large.
- **Collector** (holes): depleted below p_{C0} near the reverse-biased junction, recovering over L_C.

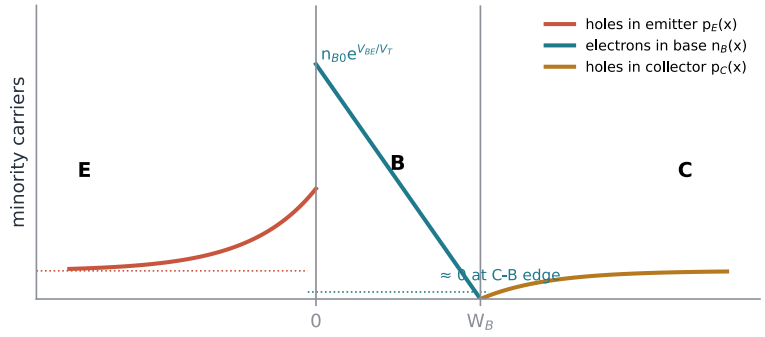


Fig 2.9 Minority carrier distributions in forward-active mode (not to scale). The slope of the base line is the collector current.

In **saturation** both junctions are forward-biased, so $n_B(W_B)$ is also large: the base fills with excess charge, which is why a saturated BJT switches off slowly. In **cut-off** both edges are ≈ 0.

2.10 The Early effect

Increase V_{CE} (so V_{CB} grows) and the C-B depletion region widens, eating into the base. The **effective base width W_B shrinks** (base-width modulation). From Fig 2.9, the same $n_B(0)$ over a shorter distance means a steeper slope, so **I_C rises** with V_{CE} instead of being flat. Extrapolate the output curves backwards and they meet near one point on the V_{CE} axis, $-V_A$, the **Early voltage**.

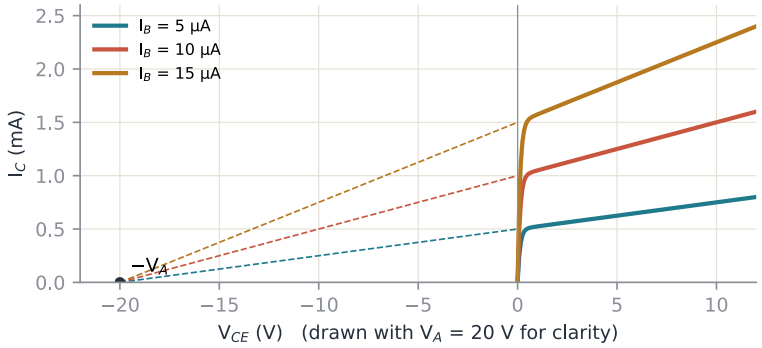


Fig 2.10 Output characteristics with base-width modulation; dashed extrapolations meet at $-V_A$.

EARLY EFFECT

$$I_C = I_S\, e^{V_{BE}/V_T}(1 + V_{CE}/V_A) \qquad r_o = \frac{V_A}{I_C} \qquad \text{intrinsic gain } g_m r_o = V_A/V_T$$

Consequences: finite output resistance (limits amplifier gain), β rising with V_{CE}, and in the extreme, **punch-through**, when the C-B depletion region reaches the E-B depletion region and the base disappears. Remedies: heavier base doping (depletion spreads less into it; but this lowers γ) or a wider base (lowers α_T); the classic BJT trade-off.

SOLVED EXAMPLE 2.10 · EARLY VOLTAGE AND R_o

At fixed I_B, a BJT gives I_C = 1.02 mA at V_{CE} = 2 V and 1.10 mA at 10 V. Find V_A, r_o, and the intrinsic gain $g_m r_o$ at about 1 mA.

- Slope = (1.10 − 1.02)/(10 − 2) = 0.01 mA/V ⇒ r_o = 1/slope = 100 kΩ.
- Line: $I_C = 1.00 + 0.01V_{CE}$ (mA) ⇒ zero at V_{CE} = −100 V ⇒ V_A = 100 V.
- $g_m r_o = V_A/V_T = 100/0.0259 \approx 3860$.

V_A = 100 V, r_o = 100 kΩ, $g_m r_o \approx 3860$

✓ Module 2: Quick revision

V_{bi} = $V_T \ln(N_a N_d/n_i^2) \approx 0.6$–$0.9$ V in Si. E_F flat in equilibrium. **Charge balance:** $N_a x_p = N_d x_n$; lighter side depletes more.

W = $\sqrt{[2\varepsilon_s(V_{bi} + V_R)(N_a + N_d)/(qN_aN_d)]}$; $W \propto \sqrt{(V_{bi} + V_R)}$.

Field: triangle, $E_{max} = qN_dx_n/\varepsilon_s = 2(V_{bi} + V_R)/W$. Potential parabolic.

Law of junction: $p_n(x_n) = p_{n0}e^{V_a/V_T}$. **Diode:** $I = I_s(e^{V/V_T} - 1)$, $I_s = Aqn_i^2(D_n/L_nN_a + D_p/L_pN_d)$; 60 mV/decade.

C_j = $\varepsilon_s A/W \propto (V_{bi} + V_R)^{-1/2}$; $1/C^2$ vs V_R linear, intercept $-V_{bi}$.

C_d = $\tau I_D/(2V_T)$ (long); $g_d = I_D/V_T$. Dominates in forward bias.

Zener: heavy doping, tunnelling, < 5 V, negative tempco. **Avalanche:** impact ionisation, > 6 V, positive tempco. **V_B** $\approx \varepsilon_s E_{crit}^2/(2qN_d) \propto 1/N_d$.

BJT: $\alpha = \gamma\alpha_T\delta$; $\beta = \alpha/(1 - \alpha)$; $\gamma \approx 1/(1 + N_BD_EW_B/N_ED_BL_E)$; $\alpha_T \approx 1 - W_B^2/2L_B^2$.

Base profile: ≈ linear from $n_{B0}e^{V_{BE}/V_T}$ to 0; $I_C = qAD_Bn_{B0}e^{V_{BE}/V_T}/W_B$.

Early: $I_C \propto (1 + V_{CE}/V_A)$; $r_o = V_A/I_C$; punch-through at the extreme.

Practice: short questions (3 marks)

1. Draw the energy band diagram of a PN junction in equilibrium and under forward bias.
2. Find V_{bi} for $N_a = N_d = 10^{16}$ cm^{-3} in Si. [0.695 V]
3. Why does the depletion region extend more into the lightly doped side?
4. Distinguish between junction capacitance and diffusion capacitance.
5. Compare Zener and avalanche breakdown (mechanism, doping, temperature coefficient).
6. Define emitter injection efficiency and base transport factor.
7. A BJT has $\alpha = 0.98$. Find β. [49]
8. What is the Early effect? How does it affect the output characteristics?

Practice: long questions (9/14 marks)

1. Using the depletion approximation, derive E(x), φ(x) and W for an abrupt junction. Sketch them.
2. Derive the ideal diode equation, stating all assumptions.
3. A p^+n Si diode has $N_d = 10^{15}$ cm^{-3}, $N_a = 10^{18}$ cm^{-3}. Find V_{bi}, W at $V_R = 10$ V and C_j for $A = 10^{-4}$ cm². [0.754 V, 3.73 µm, 0.278 pF]
4. Derive the expression for junction capacitance. Explain how a $1/C^2$ plot gives N_d and V_{bi}.
5. Explain npn current components; derive γ and α_T and show how they set β.
6. Sketch BJT minority carrier distributions in active, saturation and cut-off; explain base-width modulation.

03

Metal–Semiconductor Contacts, MOS Capacitor & MOSFET

> *The MOSFET is the device of the century: billions sit on every chip. "Derive V_T", "draw the C-V curve" and "explain the square law" come up in every VLSI and analog interview.*

SYLLABUS CHECKLIST

- ☐ Metal–semiconductor contacts: Schottky barrier and ohmic contact (band diagrams)
- ☐ Schottky diode vs PN diode
- ☐ MOS capacitor: energy band diagrams in accumulation, depletion, inversion
- ☐ C-V characteristics (low and high frequency)
- ☐ Flat-band voltage; threshold voltage derivation; body effect
- ☐ MOSFET: structure, operation
- ☐ I-V characteristics derivation (linear and saturation, square law); transconductance

11 hours · CO3 · Analyse metal-semiconductor and MOS structures and MOSFET characteristics

3.1 The Schottky barrier

Every chip needs metal wires touching semiconductor. What happens at that contact depends on one comparison: the **work function** of the metal, $q\varphi_m$ (energy to pull an electron from E_F to the vacuum level), against that of the semiconductor, $q\varphi_s = q\chi + (E_C - E_F)$, where χ is the **electron affinity** (vacuum level to E_C).

Take **n-type Si with $\varphi_m > \varphi_s$**. Before contact, the semiconductor's E_F is higher, so when they touch, electrons flow from the semiconductor into the metal until the Fermi levels line up. That leaves behind uncovered donor ions: a depletion region in the semiconductor, exactly like the n side of a p^+n junction, and the bands bend **up** towards the interface.

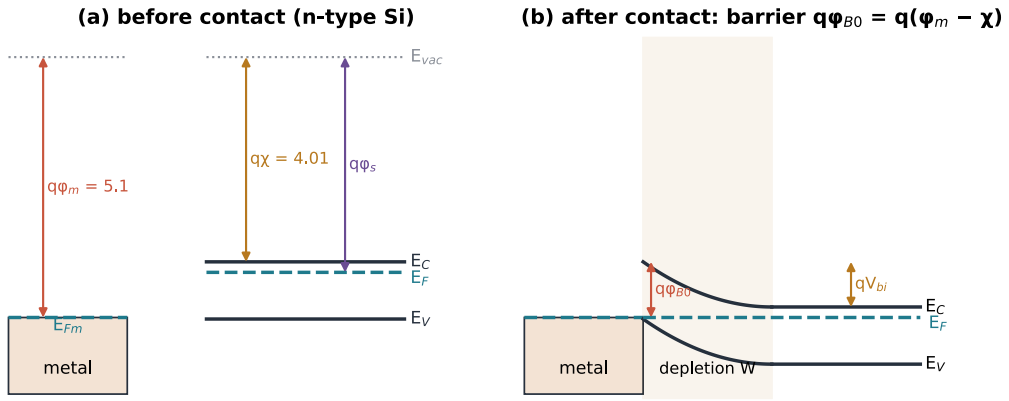


Fig 3.1 Gold ($q\varphi_m$ = 5.1 eV) on n-Si ($N_d = 10^{16}$), roughly to scale. After contact E_F is flat and a barrier $q\varphi_{B0}$ blocks electrons from the metal.

IDEAL SCHOTTKY BARRIER (N-TYPE)

$$\varphi_{B0} = \varphi_m - \chi \qquad V_{bi} = \varphi_{B0} - \varphi_n, \quad \varphi_n = (E_C - E_F)/q = V_T \ln(N_c/N_d)$$

$$W = \sqrt{[2\varepsilon_s(V_{bi} + V_R)/(qN_d)]} \qquad C' = \varepsilon_s/W \quad \text{(same as a one-sided junction)}$$

Two barriers matter: electrons in the metal see a fixed barrier φ_{B0} (independent of bias); electrons in the semiconductor see V_{bi}, which bias raises or lowers. Forward bias (metal +) lowers V_{bi} and electrons pour over into the metal by **thermionic emission**:

SCHOTTKY DIODE CURRENT (THERMIONIC EMISSION)

$$J = J_{sT}(e^{V_a/V_T} - 1) \qquad J_{sT} = A^*T^2\, e^{-\varphi_B/V_T} \quad (A^* = \text{Richardson constant} \approx 120\ \text{A/cm}^2\text{K}^2 \text{ for n-Si})$$

COMMON MISTAKE

Trusting $\varphi_{B0} = \varphi_m - \chi$ blindly. Real barriers are set largely by **interface states** that pin the Fermi level, and by image-force lowering, so measured Si barriers sit around 0.5–0.9 eV almost regardless of the metal. Use the ideal formula unless the question gives a measured barrier.

SOLVED EXAMPLE 3.1 · IDEAL SCHOTTKY CONTACT

Gold (φ_m = 5.1 V) contacts n-Si with $N_d = 10^{16}$ cm^{-3} (χ = 4.01 V). Find φ_{B0}, V_{bi}, W and E_{max} at zero bias.

- $\varphi_{B0} = 5.1 - 4.01 = 1.09$ V. $\varphi_n = 0.0259 \ln(2.8 \times 10^{19}/10^{16}) = 0.206$ V.
- $V_{bi} = 1.09 - 0.206 = 0.884$ V.
- $W = \sqrt{(2 \times 1.036 \times 10^{-12} \times 0.884/(1.602 \times 10^{-19} \times 10^{16}))} = 3.38 \times 10^{-5}$ cm = 0.338 µm.
- $E_{max} = qN_dW/\varepsilon_s = 5.23 \times 10^4$ V/cm.

φ_{B0} = 1.09 V, V_{bi} = 0.884 V, W = 0.338 µm, $E_{max} = 5.23 \times 10^4$ V/cm

3.2 Ohmic contacts

An **ohmic contact** passes current equally easily in both directions with negligible resistance: a linear, low-value I-V. Every device needs at least two. There are two ways to make one:

- **No barrier:** on n-type choose $\varphi_m < \varphi_s$. Electrons flow from the metal into the semiconductor, the surface is **accumulated**, and electrons move freely either way (Fig 3.2a). In practice suitable metals are hard to find for Si because of Fermi-level pinning.
- **Tunnelling (the practical method):** dope the semiconductor very heavily (n^+, $\gtrsim 10^{19}$ cm^{-3}) under the contact. A barrier still exists, but $W \propto 1/\sqrt{N_d}$ becomes only a few nanometres thick and electrons **tunnel** straight through (Fig 3.2b). This is why MOSFET source/drain regions are n^+, and why contacts are silicided.

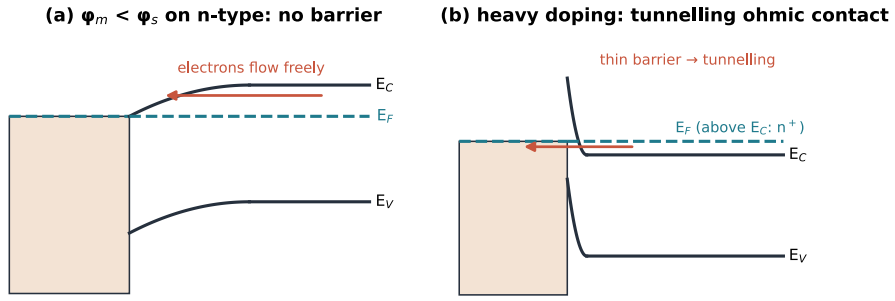


Fig 3.2 Two routes to an ohmic contact on n-type silicon.

SOLVED EXAMPLE 3.2 · WHY HEAVY DOPING MAKES CONTACTS OHMIC

A 0.7 V barrier sits on n-Si. Find the depletion width for $N_d = 10^{17}$, 10^{19} and 10^{20} cm^{-3}. Tunnelling becomes strong below about 10 nm.

- $W = \sqrt{(2\varepsilon_s V/(qN_d))}$ with $2\varepsilon_s V/q = 9.05 \times 10^6$ cm^{-1}.
- 10^{17}: 95 nm (rectifying). 10^{19}: 9.5 nm (tunnelling begins). 10^{20}: 3.0 nm (strongly ohmic).

95 nm → 9.5 nm → 3.0 nm: heavy doping thins the barrier enough to tunnel

3.3 Schottky diode vs PN diode

Both rectify, but the physics differs. A PN diode's current is **minority carriers** diffusing in the neutral regions. A Schottky diode's current is **majority carriers** (electrons) hopping over a barrier into the metal, where they are instantly just more electrons. No minority charge is stored.

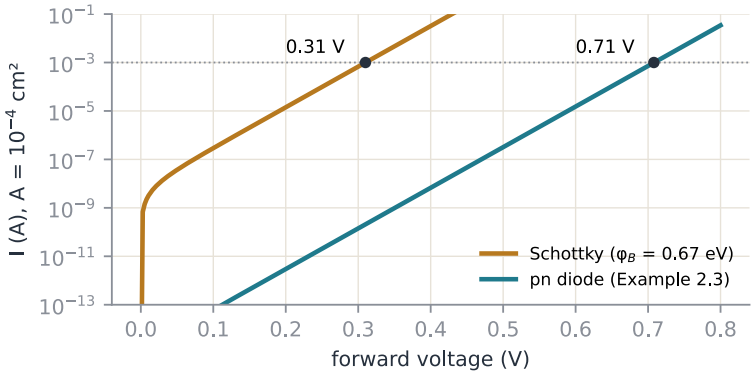


Fig 3.3 Computed I-V of a Schottky diode (φ_B = 0.67 eV) and the PN diode of Example 2.3, same area. The Schottky turns on about 0.4 V earlier.

Property	Schottky diode	PN diode
Current carriers	Majority (thermionic emission)	Minority (diffusion)
Saturation current	Large (≈ 10^{-5} A/cm²)	Tiny (≈ 10^{-11} A/cm²)
Turn-on voltage	≈ 0.2–0.4 V	≈ 0.6–0.7 V
Stored charge / reverse recovery	Negligible: very fast (GHz, switching supplies)	Diffusion capacitance, slow recovery
Reverse leakage, breakdown	Higher leakage, lower breakdown	Lower leakage, higher breakdown
Typical use	RF mixers, clamps (Schottky TTL), SMPS rectifiers	General rectification, high voltage

SOLVED EXAMPLE 3.3 · TURN-ON COMPARISON INTERVIEW

A Schottky diode has φ_B = 0.67 V, A* = 120 A/cm²K², A = 10^{-4} cm². Find J_{sT} at 300 K and the voltage for 1 mA. Compare with the PN diode of Example 2.3 ($J_s = 1.32 \times 10^{-11}$ A/cm²).

- $J_{sT} = 120 \times 300^2 \times e^{-0.67/0.0259} = 1.08 \times 10^7 \times 5.83 \times 10^{-12} = 6.29 \times 10^{-5}$ A/cm².
- $I_s = 6.29 \times 10^{-9}$ A ⇒ $V(1\text{ mA}) = 0.0259 \ln(10^{-3}/6.29 \times 10^{-9}) = 0.310$ V.
- Ratio of saturation currents = 4.8×10^6 ⇒ $\Delta V = V_T \ln(4.8 \times 10^6) = 0.398$ V (PN needs 0.708 V).

$J_{sT} = 6.29 \times 10^{-5}$ A/cm²; 0.31 V vs 0.71 V for 1 mA (≈ 0.4 V lower)

3.4 MOS capacitor: band diagrams

A MOS capacitor is a sandwich: **M**etal (or heavily doped polysilicon) gate, a thin **O**xide insulator (SiO_2), and a **S**emiconductor body, here p-type. No DC current flows through the oxide, so the semiconductor Fermi level stays flat; the gate voltage V_G just shifts the gate's E_F by $-qV_G$ and bends the semiconductor bands near the surface.

THINK OF IT LIKE

A capacitor whose bottom plate is "soft". Put + charge on the gate and the semiconductor must supply equal − charge. First it pushes holes away (exposing − acceptor ions: depletion). Push harder and it finally pulls in free electrons right at the surface (inversion): the p-type surface has become n-type.

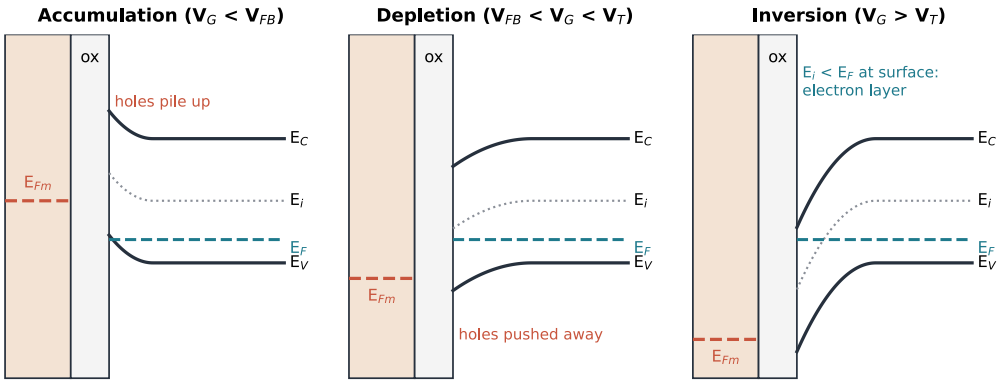


Fig 3.4 p-substrate MOS capacitor in its three regimes. Positive V_G lowers E_{Fm} and bends the bands down at the surface.

Regime	Gate voltage	Band bending (surface potential ψ_s)	Surface charge
Accumulation	$V_G < V_{FB}$	Up ($\psi_s < 0$)	Extra holes (majority) pile up
Flat band	$V_G = V_{FB}$	None ($\psi_s = 0$)	Neutral
Depletion	$V_{FB} < V_G < V_T$	Down, $0 < \psi_s < 2\varphi_{fp}$	Ionised acceptors $-qN_a x_d$
Inversion	$V_G > V_T$	Down, $\psi_s \approx 2\varphi_{fp}$ (pinned)	Electron layer Q_n + fixed depletion charge

Threshold (strong inversion) condition. Define $\varphi_{fp} = (E_i - E_F)/q = V_T \ln(N_a/n_i)$ in the bulk. When the surface has bent by $\psi_s = 2\varphi_{fp}$, the surface is as strongly n-type as the bulk is p-type ($n_{surface} = N_a$). Beyond this, the electron charge grows exponentially with ψ_s, so ψ_s barely moves and the depletion width stops at its maximum:

MAXIMUM DEPLETION WIDTH

$$x_{dT} = \sqrt{\left[\frac{4\varepsilon_s \varphi_{fp}}{qN_a}\right]} \qquad |Q'_{SD(max)}| = qN_a x_{dT}$$

3.5 C-V characteristics

Measure the small-signal capacitance $C = dQ_G/dV_G$ while sweeping V_G. The MOS structure behaves like C_{ox} in series with the semiconductor's own capacitance C_s: $1/C = 1/C_{ox} + 1/C_s$, with $C_{ox} = \varepsilon_{ox}/t_{ox}$ per unit area.

- **Accumulation:** charge changes right at the surface, C_s is huge ⇒ $C \approx C_{ox}$.
- **Depletion:** charge changes at the edge of the depletion region, $C_s = \varepsilon_s/x_d$; as x_d grows, C falls.
- **Inversion, low frequency** ($\lesssim$ 10–100 Hz): minority electrons can be generated fast enough to follow the signal, so charge again changes at the surface ⇒ C climbs back to C_{ox}.
- **Inversion, high frequency** ($\approx$ 1 MHz): generation is too slow; the inversion charge cannot follow, the depletion edge (fixed at x_{dT}) responds instead ⇒ C stays at its minimum.

C-V KEY VALUES

$$C_{ox} = \varepsilon_{ox}/t_{ox} \qquad C'_{min} = \frac{\varepsilon_{ox}}{t_{ox} + (\varepsilon_{ox}/\varepsilon_s)x_{dT}} \qquad C'_{FB} = \frac{\varepsilon_{ox}}{t_{ox} + (\varepsilon_{ox}/\varepsilon_s)L_D}, \quad L_D = \sqrt{(\varepsilon_s V_T/qN_a)}$$

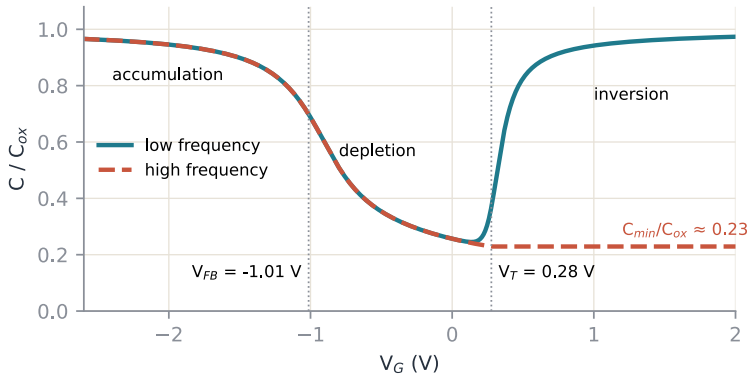


Fig 3.5 C-V of the Example 3.4 capacitor ($N_a = 10^{17}$, t_{ox} = 10 nm), computed from the exact charge equation. The full model gives C_{min}/C_{ox} = 0.23; the depletion approximation in Example 3.5 gives 0.226.

EXAM TIP

From a measured HF C-V curve you can read t_{ox} (from $C_{max} = C_{ox}$), N_a (from C_{min}/C_{ox}), V_{FB} and hence oxide charge. Fixed positive oxide charge shifts the whole curve **left** (more negative); the shape does not change. For an n-type substrate the curve is mirror-imaged (accumulation at positive V_G).

3.6 Flat-band voltage

With $V_G = 0$, are the bands flat? Usually not, for two reasons: (1) the gate and the semiconductor have different work functions, so their Fermi levels would not line up without bending; (2) the oxide contains trapped positive charge Q'_{ss} (fixed charge at the Si/SiO_2 interface), which induces negative charge in the silicon. The gate voltage that undoes both and makes the bands flat is the **flat-band voltage**:

FLAT-BAND VOLTAGE

$$V_{FB} = \varphi_{ms} - \frac{Q'_{ss}}{C_{ox}} \qquad \varphi_{ms} = \varphi_m - (\chi + E_g/2q + \varphi_{fp}) \ \text{(p-substrate)}$$

n^+ poly gate ($E_F \approx E_C$, $\varphi_m = \chi$): $\varphi_{ms} = -(E_g/2q + \varphi_{fp})$. p^+ poly gate: $\varphi_{ms} = +(E_g/2q - \varphi_{fp})$.

3.7 Threshold voltage derivation

Threshold is the gate voltage that brings the surface to $\psi_s = 2\varphi_{fp}$. Follow the charge and the voltages:

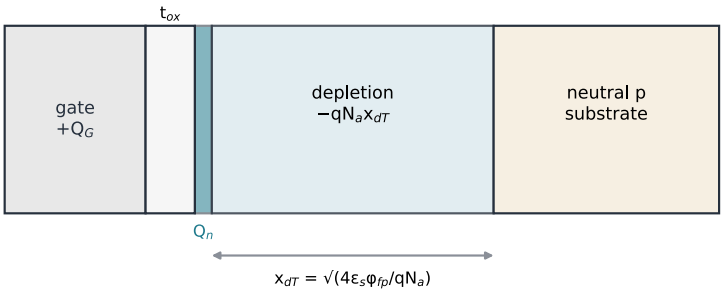


Fig 3.6 Charges at threshold: gate charge balances the maximum depletion charge (inversion charge Q_n is just starting).

Charge neutrality at threshold: $Q'_G + Q'_{ss} = |Q'_{SD(max)}| = qN_a x_{dT}$ ($Q_n \approx 0$ at the onset).
Voltage around the loop: $V_G = V_{ox} + \psi_s + \varphi_{ms}$, with $V_{ox} = Q'_G/C_{ox}$ (Gauss's law across the oxide).
Substitute $\psi_s = 2\varphi_{fp}$ and $Q'_G = |Q'_{SD(max)}| - Q'_{ss}$, then group $\varphi_{ms} - Q'_{ss}/C_{ox}$ as V_{FB}.

THRESHOLD VOLTAGE (N-CHANNEL, P-SUBSTRATE)

$$V_{TN} = V_{FB} + 2\varphi_{fp} + \frac{|Q'_{SD(max)}|}{C_{ox}} = \varphi_{ms} - \frac{Q'_{ss}}{C_{ox}} + 2\varphi_{fp} + \frac{\sqrt{(4q\varepsilon_s N_a \varphi_{fp})}}{C_{ox}}$$

p-channel (n-substrate): $V_{TP} = \varphi_{ms} - Q'_{ss}/C_{ox} - 2\varphi_{fn} - |Q'_{SD(max)}|/C_{ox}$.

Read it as a recipe: *first flatten the bands* (V_{FB}), *then bend the surface by* $2\varphi_{fp}$, *then support the depletion charge across the oxide*. The knobs a process engineer turns: substrate doping (threshold-adjust implant), oxide thickness, and gate material.

SOLVED EXAMPLE 3.4 · THRESHOLD VOLTAGE KTU FAVOURITE

An n^+ poly-gate nMOS has $N_a = 10^{17}$ cm^{-3}, $t_{ox} = 10$ nm, $Q'_{ss} = 10^{11}$ q C/cm². Find C_{ox}, φ_{fp}, x_{dT}, V_{FB} and V_T. What happens to V_T if $N_a = 10^{16}$?

- $C_{ox} = 3.453 \times 10^{-13}/10^{-6} = 3.45 \times 10^{-7}$ F/cm². $\varphi_{fp} = 0.0259 \ln(10^{17}/1.5 \times 10^{10}) = 0.407$ V.
- $x_{dT} = \sqrt{(4 \times 1.036 \times 10^{-12} \times 0.407/(1.602 \times 10^{-19} \times 10^{17}))} = 1.03 \times 10^{-5}$ cm = 0.103 µm.
- $|Q'_{SD}| = qN_a x_{dT} = 1.64 \times 10^{-7}$ C/cm² ⇒ $|Q'_{SD}|/C_{ox} = 0.476$ V.
- $\varphi_{ms} = -(0.56 + 0.407) = -0.967$ V. $Q'_{ss}/C_{ox} = 1.602 \times 10^{-8}/3.45 \times 10^{-7} = 0.046$ V ⇒ $V_{FB} = -1.013$ V.
- $V_T = -1.013 + 0.814 + 0.476 = 0.277$ V.
- With $N_a = 10^{16}$: $\varphi_{fp} = 0.347$, $x_{dT} = 0.300$ µm, $Q/C_{ox} = 0.139$, $V_{FB} = -0.954$ ⇒ $V_T = -0.954 + 0.695 + 0.139 = -0.12$ V. The transistor would already be ON at $V_{GS} = 0$ (depletion-mode).

$C_{ox} = 345$ nF/cm², $\varphi_{fp} = 0.407$ V, $x_{dT} = 0.103$ µm, $V_{FB} = -1.01$ V, $V_T = +0.277$ V (-0.12 V at 10^{16})

SOLVED EXAMPLE 3.5 · MINIMUM CAPACITANCE

For the capacitor of Example 3.4, find C'_{min} and C_{min}/C_{ox} for the high-frequency C-V curve.

- $(\varepsilon_{ox}/\varepsilon_s)x_{dT} = (3.9/11.7) \times 102.6$ nm = 34.2 nm.
- $C'_{min} = 3.453 \times 10^{-13}/(10 + 34.2) \times 10^{-7} = 7.81 \times 10^{-8}$ F/cm². Ratio = 10/44.2 = 0.226.

$C'_{min} = 78.1$ nF/cm², $C_{min}/C_{ox} = 0.226$

3.8 The body effect

In a real circuit the source of an nMOS is not always at the body (substrate) potential: in a stack of transistors the upper ones have their sources above ground. A positive source-to-body voltage V_{SB} reverse-biases the source and channel with respect to the body. The surface must now bend by $2\varphi_{fp} + V_{SB}$ to invert, the depletion region gets wider, more depletion charge must be supported, and **V_T rises**.

BODY EFFECT

$$V_T = V_{T0} + \gamma\left(\sqrt{(2\varphi_{fp} + V_{SB})} - \sqrt{(2\varphi_{fp})}\right) \qquad \gamma = \frac{\sqrt{(2q\varepsilon_s N_a)}}{C_{ox}} \text{ (body-effect coefficient, V}^{1/2}\text{)}$$

SOLVED EXAMPLE 3.6 · BODY EFFECT

For the device of Example 3.4 ($V_{T0} = 0.277$ V), find γ and V_T at $V_{SB} = 2$ V.

- $\gamma = \sqrt{(2 \times 1.602 \times 10^{-19} \times 1.036 \times 10^{-12} \times 10^{17})}/3.45 \times 10^{-7} = 0.528$ V$^{1/2}$.

▸ $\Delta V_T = 0.528(\sqrt{2.814} - \sqrt{0.814}) = 0.528(1.677 - 0.902) = 0.409$ V.

$\gamma = 0.53\ V^{1/2}$; $V_T = 0.277 + 0.409 = 0.686$ V

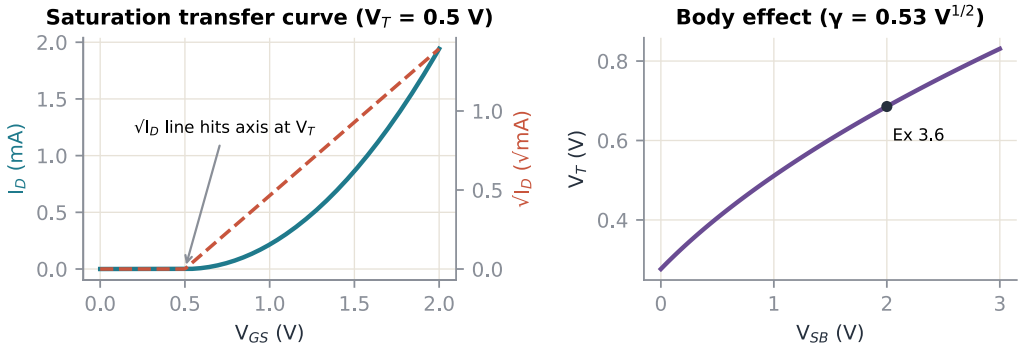


Fig 3.7 Left: saturation transfer curve of the Example 3.7 device; $\sqrt{I_D}$ is a straight line whose intercept is V_T. Right: V_T vs V_{SB} for Example 3.6.

INTERVIEW CORNER

Q. Why does the HF C-V curve stay flat in inversion?

A. Minority carriers are supplied only by slow thermal generation; at 1 MHz they cannot follow the signal, so only the depletion edge responds and C stays at C_{min}. In a MOSFET the source supplies electrons quickly, so the gate C-V looks like the LF curve.

Q. How do you raise V_T of an nMOS?

A. Higher substrate doping (threshold-adjust implant), thicker oxide, a gate with higher work function (p^+ poly or a suitable metal), or back-bias ($V_{SB} > 0$).

Q. Why did industry move from poly gates to metal gates with high-k oxides?

A. Thin SiO_2 leaked by tunnelling; high-k (HfO_2) gives the same C_{ox} with a thicker film. Poly gates also suffer gate depletion and Fermi-level pinning on high-k, so metal gates with tuned work functions are used to set V_T.

Q. Why does current not drop to zero after pinch-off?

A. Pinch-off is not a blockage: the depleted gap near the drain has a strong lateral field that sweeps arriving electrons to the drain, like a reverse-biased collector junction. The current is set by how many electrons the channel delivers.

Q. Why is the pMOS made wider than the nMOS in a CMOS inverter?

A. Hole mobility is about 2–3× lower, so the pMOS needs a proportionally larger W/L to match the nMOS current and give symmetric rise and fall times.

Q. Is the square law valid in modern devices?

A. Only roughly. In short channels velocity saturation makes I_D closer to linear in V_{ov}, and mobility degrades with vertical field (Module 4).

3.9 MOSFET structure and operation

Add two n^+ regions, the **source** and **drain**, on either side of the MOS capacitor's gate and you have an n-channel MOSFET (nMOS). Channel length L is the gate length between them; width W is into the page. The body (substrate) is p-type and usually tied to the lowest potential.

- **$V_{GS} < V_T$ (cut-off):** source and drain are two back-to-back PN junctions with p-type silicon between them. Essentially no current (apart from the subthreshold current of Module 4).
- **$V_{GS} > V_T$, small V_{DS} (linear/triode):** an inversion layer of electrons connects source to drain. It behaves like a resistor whose conductance is set by the gate: a voltage-controlled resistor.
- **Larger V_{DS}:** the channel is less inverted near the drain (gate-to-channel voltage there is only $V_{GS} - V_{DS}$), so it tapers and the I-V curve bends over.
- **$V_{DS} \geq V_{GS} - V_T$ (saturation):** the channel **pinches off** at the drain end. Electrons reaching the pinch-off point are swept across the short depleted gap by the high field, just as the collector junction collects in a BJT. Current stops rising with V_{DS}.

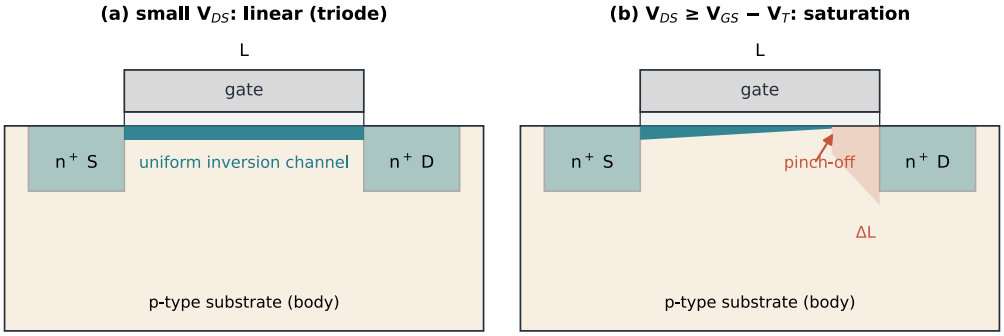


Fig 3.8 nMOS cross-section: uniform channel at small V_{DS}; pinched-off channel in saturation (ΔL is the pinched region, see channel length modulation in Module 4).

THINK OF IT LIKE

A garden hose (the channel) whose wall softness is the gate voltage. Squeeze the far end by raising the drain voltage and the flow increases at first, then the end pinches nearly shut. The flow is now set by how much water the hose delivers to the pinch point, not by how hard you pull from the far end.

3.10 Derivation of the MOSFET I-V characteristics

Assumptions (the long-channel, "gradual channel" model): (1) current is only drift of electrons along the channel (y direction); (2) the vertical field (from the gate) is much larger than the lateral field, so each slice of the channel behaves like a MOS capacitor with local channel voltage V(y); (3) constant mobility μ_n; (4) depletion charge variation along the channel ignored; (5) $V_{SB} = 0$.

At position y the channel voltage is V(y) (0 at source, V_{DS} at drain). Inversion charge per area there: $Q'_n(y) = -C_{ox}[V_{GS} - V_T - V(y)]$.

Drift current through the channel cross-section: $I_D = -W\,Q'_n(y)\,\mu_n\,dV/dy$ (electron velocity $\mu_n dV/dy$, charge per length WQ'_n). I_D is the same at every y.
Separate variables: $I_D\,dy = \mu_n C_{ox} W\,[V_{GS} - V_T - V]\,dV$.
Integrate y from 0 to L and V from 0 to V_{DS}: $I_D L = \mu_n C_{ox} W[(V_{GS} - V_T)V_{DS} - V_{DS}^2/2]$.
This parabola peaks where $dI_D/dV_{DS} = 0$, i.e. $V_{DS(sat)} = V_{GS} - V_T$: exactly where $Q'_n(L) = 0$ (pinch-off). Beyond it the current stays at the peak value.

LONG-CHANNEL NMOS (SQUARE LAW)

$$\text{Linear } (V_{DS} < V_{GS} - V_T): \quad I_D = \mu_n C_{ox} \frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right]$$

$$\text{Saturation } (V_{DS} \geq V_{GS} - V_T): \quad I_D = \frac{1}{2}\mu_n C_{ox} \frac{W}{L}(V_{GS} - V_T)^2$$

Deep triode ($V_{DS} \ll V_{ov}$): $R_{on} = 1/[\mu_n C_{ox}(W/L)(V_{GS} - V_T)]$. Overdrive $V_{ov} = V_{GS} - V_T$; process transconductance $k'_n = \mu_n C_{ox}$.

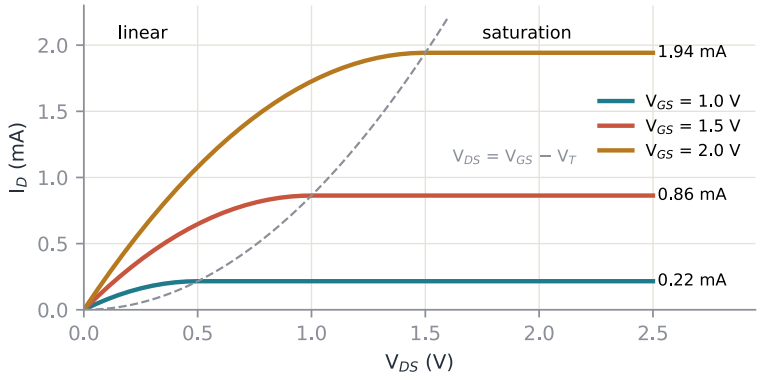


Fig 3.9 Output characteristics of the Example 3.7 device (k'_nW/L = 1.73 mA/V², V_T = 0.5 V). The dashed parabola separates linear and saturation regions.

COMMON MISTAKE

Calling the flat region "saturation" and thinking it means the same as BJT saturation. They are opposites: MOSFET saturation (flat current) corresponds to the BJT's *active* region; the MOSFET's triode region is where it acts like a BJT in saturation (a closed switch).

SOLVED EXAMPLE 3.7 · DRAIN CURRENT FREQUENTLY ASKED

An nMOS has μ_n = 500 cm²/V·s, t_{ox} = 10 nm, W/L = 10, V_T = 0.5 V. With V_{GS} = 1.5 V find k'_n, I_D in saturation, I_D at V_{DS} = 0.2 V, and R_{on} in deep triode.

- $k'_n = \mu_n C_{ox} = 500 \times 3.453 \times 10^{-7} = 172.7\ \mu A/V^2$; $k'_nW/L = 1.727$ mA/V². $V_{ov} = 1$ V.
- Saturation: $I_D = ½ \times 1.727 \times 1^2 = 0.863$ mA (valid for $V_{DS} \geq 1$ V).
- $V_{DS} = 0.2$ V < 1 V ⇒ linear: $I_D = 1.727 \times (1 \times 0.2 - 0.02) = 0.311$ mA.
- $R_{on} = 1/(1.727 \times 10^{-3} \times 1) = 579\ \Omega$.

k'_n = 172.7 µA/V²; $I_{D,sat}$ = 0.863 mA; I_D(0.2 V) = 0.311 mA; R_{on} = 579 Ω

SOLVED EXAMPLE 3.8 · WHICH REGION? GATE

An nMOS with V_T = 0.7 V and $k'_n(W/L)$ = 0.4 mA/V² has V_{GS} = 2 V. Find I_D for V_{DS} = 1 V and V_{DS} = 2 V.

- V_{ov} = 1.3 V. V_{DS} = 1 V < 1.3 V ⇒ linear: I_D = 0.4(1.3 × 1 − 0.5) = 0.32 mA.
- V_{DS} = 2 V > 1.3 V ⇒ saturation: I_D = 0.2 × 1.3^2 = 0.338 mA.

0.320 mA (linear) and 0.338 mA (saturation)

3.11 Transconductance

Transconductance $g_m = \partial I_D/\partial V_{GS}$ (at constant V_{DS}) says how strongly the gate controls the current: the "gain" of the device. Differentiating the two I-V equations:

TRANSCONDUCTANCE

$$\text{Linear: } g_m = \mu_n C_{ox}\frac{W}{L}V_{DS} \qquad \text{Saturation: } g_m = \mu_n C_{ox}\frac{W}{L}(V_{GS}-V_T) = \frac{2I_D}{V_{ov}} = \sqrt{(2\mu_n C_{ox}(W/L)I_D)}$$

To increase g_m: larger W/L, higher mobility (nMOS beats pMOS by ≈ 2–3×), thinner oxide, more current. Compare a BJT: $g_m = I_C/V_T$. At the same current a BJT gives more g_m, because $V_{ov}/2$ (typically 0.1–0.2 V) is larger than V_T = 26 mV.

SOLVED EXAMPLE 3.9 · G_M AND SIZING

(a) Find g_m for the Example 3.7 device in saturation. (b) Using the same process (k'_n = 172.7 µA/V²), choose W/L to give I_D = 0.5 mA at V_{ov} = 0.25 V, and find g_m.

- (a) g_m = 1.727 mA/V² × 1 V = 1.73 mS (check: 2 × 0.863/1 = 1.73 mS).
- (b) $W/L = 2I_D/(k'_n V_{ov}^2) = 10^{-3}/(1.727 \times 10^{-4} \times 0.0625)$ = 92.7. $g_m = 2I_D/V_{ov}$ = 4 mS.

(a) 1.73 mS · (b) W/L ≈ 93, g_m = 4 mS

✓ Module 3: Quick revision

Schottky (n-type, $\varphi_m > \varphi_s$): $\varphi_{B0} = \varphi_m - \chi$; $V_{bi} = \varphi_{B0} - \varphi_n$; W like one-sided junction.

Thermionic: $J_{sT} = A^*T^2e^{-\varphi_B/V_T}$; majority carriers, no stored charge, ≈ 0.3 V turn-on.

Ohmic: $\varphi_m < \varphi_s$ (n-type) or heavy doping ⇒ thin barrier, tunnelling.

MOS regimes (p-sub): accumulation ($V_G < V_{FB}$), depletion, inversion ($\psi_s = 2\varphi_{fp}$).

φ_{fp} $= V_T \ln(N_a/n_i)$; $x_{dT} = \sqrt{(4\varepsilon_s\varphi_{fp}/qN_a)}$; $Q'_{SD} = qN_ax_{dT}$.

C-V: $1/C = 1/C_{ox} + 1/C_s$. LF returns to C_{ox} in inversion; HF stays at $C_{min} = \varepsilon_{ox}/(t_{ox} + (\varepsilon_{ox}/\varepsilon_s)x_{dT})$.

V_{FB} $= \varphi_{ms} - Q'_{ss}/C_{ox}$; n^+ poly on p: $\varphi_{ms} = -(E_g/2q + \varphi_{fp})$.

V_{TN} $= V_{FB} + 2\varphi_{fp} + \sqrt{(4q\varepsilon_sN_a\varphi_{fp})}/C_{ox}$.

Body effect: $\Delta V_T = \gamma(\sqrt{(2\varphi_{fp} + V_{SB})} - \sqrt{(2\varphi_{fp})})$, $\gamma = \sqrt{(2q\varepsilon_sN_a)}/C_{ox}$.

Linear: $I_D = k'_n(W/L)[V_{ov}V_{DS} - V_{DS}^2/2]$; $R_{on} = 1/(k'_n(W/L)V_{ov})$.

Saturation: $V_{DS} \geq V_{ov}$; $I_D = ½k'_n(W/L)V_{ov}^2$; pinch-off at the drain.

g_m $= k'_n(W/L)V_{ov} = 2I_D/V_{ov} = \sqrt{(2k'_n(W/L)I_D)}$; linear region $g_m = k'_n(W/L)V_{DS}$.

Practice: short questions (3 marks)

1. Draw the band diagram of a Schottky contact on n-Si before and after contact.
2. How is an ohmic contact made on lightly doped Si?
3. Give four differences between a Schottky diode and a PN diode.
4. Draw MOS band diagrams for accumulation and inversion (p-substrate).
5. Why do LF and HF C-V curves differ in inversion?
6. Define flat-band voltage. What shifts it?
7. Find φ_{fp} and x_{dT} for $N_a = 10^{16}$ cm^{-3}. [0.347 V, 0.30 µm]
8. Define transconductance; give its saturation-region expression.

Practice: long questions (9/14 marks)

1. Explain Schottky barrier formation with band diagrams; derive V_{bi} and W.
2. Explain LF and HF C-V characteristics of a p-substrate MOS capacitor.
3. Derive the threshold voltage of an n-channel MOSFET. Find V_T for $N_a = 3 \times 10^{16}$, $t_{ox} = 20$ nm, n^+ poly, $Q'_{ss} = 0$. [0.32 V]
4. Explain the body effect and derive ΔV_T. Evaluate γ for $N_a = 10^{16}$, $t_{ox} = 10$ nm. [0.167 V$^{1/2}$]
5. Derive MOSFET linear and saturation I-V (gradual channel approximation).
6. A MOSFET has $k'_n = 100$ µA/V², $W/L = 20$, $V_T = 0.6$ V, $V_{GS} = 1.6$ V. Find I_D at $V_{DS} = 0.5$ V and 2 V, and g_m in saturation. [0.75 mA, 1.0 mA, 2 mS]

04

Scaling, Short-Channel Effects, MESFET & FinFET

> *Moore's law is the story of shrinking the MOSFET, and every problem in this module is a problem that chip designers at Intel, TI or Qualcomm fight daily: leakage, DIBL, velocity saturation, reliability. FinFET questions are now standard in VLSI interviews.*

SYLLABUS CHECKLIST

- ☐ MOSFET scaling: constant-field and constant-voltage
- ☐ Subthreshold conduction and subthreshold swing
- ☐ Short-channel effects: channel length modulation, DIBL, velocity saturation
- ☐ Hot-carrier effects; threshold voltage roll-off
- ☐ MESFET: structure and operation
- ☐ FinFET: structure, and why it beats the planar MOSFET

8 hours · CO4 · Explain scaling, short-channel effects and advanced FET structures

4.1 MOSFET scaling

Why shrink transistors? Smaller devices mean more of them per chip, lower capacitance, faster switching and lower energy per operation. But you cannot just shrink L and keep everything else: fields would rise, junctions would break down and the gate would lose control. **Scaling rules** tell us how to shrink all dimensions and voltages together by a factor $\kappa > 1$ ($\kappa \approx 1.4$ per technology node, which halves the area).

Constant-field (Dennard) scaling

Shrink every dimension (L, W, t_{ox}, junction depth) by κ, reduce all voltages by κ, and raise doping by κ. Then every internal electric field stays the same, so the physics inside the device is unchanged, just smaller.

Quantity	Constant-field	Constant-voltage
Dimensions L, W, t_{ox}, x_j	$1/\kappa$	$1/\kappa$
Supply V_{DD}, V_T	$1/\kappa$	1
Doping N_a	κ	κ^2
Electric field	1	κ
C_{ox} per area / gate capacitance	κ / $1/\kappa$	κ / $1/\kappa$
Drain current I_D	$1/\kappa$	κ
Gate delay CV/I	$1/\kappa$	$1/\kappa^2$
Power per device VI	$1/\kappa^2$	κ
Device density	κ^2	κ^2
Power density	**1**	**κ^3**
Energy per switch (power-delay product)	$1/\kappa^3$	$1/\kappa$

Constant-voltage scaling keeps V_{DD} fixed (compatible with older chips), which was done in the 1970s–80s, but fields grow by κ, causing hot-carrier and oxide breakdown problems, and power density explodes as κ^3. Constant-field scaling is ideal but V_T cannot really scale (subthreshold leakage, next section), so modern scaling is a compromise ("generalised scaling"). Around 2005 voltage scaling stalled and the "free" speed-up ended: that is why clock speeds stopped near 3–5 GHz and chips went multi-core.

SOLVED EXAMPLE 4.1 · SCALING BY K = 2

A MOSFET with L = 1 µm, t_{ox} = 20 nm, V_{DD} = 5 V is scaled by κ = 2. Give the new L, t_{ox}, V_{DD} and the factors for current, delay, power per device and power density under (a) constant-field and (b) constant-voltage scaling.

▸ (a) L = 0.5 µm, t_{ox} = 10 nm, V_{DD} = 2.5 V, N_a × 2. I_D ÷ 2, delay ÷ 2, power ÷ 4, density × 4 ⇒ power density unchanged; energy per switch ÷ 8.

▸ (b) L = 0.5 µm, t_{ox} = 10 nm, V_{DD} = 5 V. Field × 2, I_D × 2, delay ÷ 4, power per device × 2, density × 4 ⇒ power density × 8.

CF: same power density, 2× faster · CV: 4× faster but 8× power density

4.2 Subthreshold conduction and subthreshold swing

The square law says $I_D = 0$ for $V_{GS} < V_T$. Not true. Below threshold the surface is in **weak inversion**: a few electrons exist at the surface, and the source-channel junction is like a forward-biased diode whose barrier the gate controls. Current flows by **diffusion** and depends **exponentially** on V_{GS}, like a BJT.

The gate does not control the surface potential perfectly: C_{ox} and the depletion capacitance C_{dep} form a divider, so the surface moves by only $\Delta V_{GS}/n$, where $n = 1 + C_{dep}/C_{ox}$.

SUBTHRESHOLD CURRENT AND SWING

$$I_D \approx I_0\, e^{(V_{GS} - V_T)/(nV_T)}\left(1 - e^{-V_{DS}/V_T}\right) \qquad n = 1 + \frac{C_{dep}}{C_{ox}}$$

$$S = \frac{dV_{GS}}{d(\log_{10} I_D)} = 2.3\,\frac{kT}{q}\left(1 + \frac{C_{dep}}{C_{ox}}\right) \geq 59.6 \text{ mV/decade at 300 K}$$

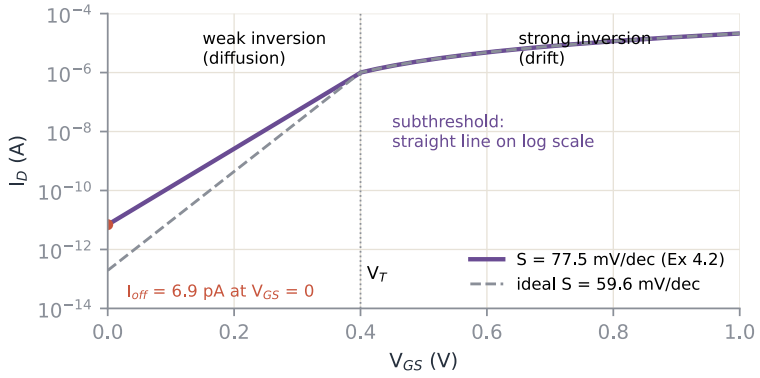


Fig 4.1 Log I_D vs V_{GS} for Example 4.2: a straight line below V_T whose slope is 1/S.

KEY IDEA

The ≈ 60 mV/decade limit is set by thermodynamics (the Boltzmann tail), not by technology. To cut the off-current by 10× you must raise V_T by at least 60 mV, which costs speed. This is the reason V_T and V_{DD} cannot keep scaling, and why better gate control (thin oxide, FinFET, GAA) that brings S closer to 60 mV/dec is so valuable.

SOLVED EXAMPLE 4.2 · OFF-CURRENT INTERVIEW

A MOSFET has $C_{dep}/C_{ox} = 0.3$, $V_T = 0.4$ V and $I_D(V_{GS} = V_T) = 1$ µA. Find S, the off-current at $V_{GS} = 0$, the off-current if S were ideal, and the V_T needed for $I_{off} = 1$ nA.

- S = 59.6 × 1.3 = 77.5 mV/dec.
- 0.4 V below V_T = 0.4/0.0775 = 5.16 decades ⇒ $I_{off} = 10^{-6} \times 10^{-5.16} = 6.9$ pA.
- Ideal S: 0.4/0.0596 = 6.71 decades ⇒ 0.20 pA.
- 1 nA is 3 decades below 1 µA ⇒ $V_T = 3 \times 77.5$ mV = 0.233 V.

S = 77.5 mV/dec; I_{off} = 6.9 pA (0.20 pA ideal); V_T = 0.23 V for 1 nA

4.3 Short-channel effects and channel length modulation

A MOSFET is "short-channel" when L becomes comparable to the depletion widths of the source and drain junctions. Then the drain starts to influence the channel directly, and the neat long-channel assumptions (gate alone controls the channel, low lateral field, constant mobility) break down. The family of resulting problems is called **short-channel effects** (sections 4.3–4.7).

Channel length modulation (CLM)

In saturation the pinch-off point moves towards the source as V_{DS} grows; the pinched-off region ΔL widens. The effective channel L − ΔL gets shorter, and since $I_D \propto 1/L$, the current keeps rising slowly: the MOSFET version of the Early effect.

CHANNEL LENGTH MODULATION

$$I_D = \frac{1}{2} k'_n \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS}) \qquad r_o = \frac{1}{\lambda I_D} \qquad \lambda \propto \frac{1}{L}$$

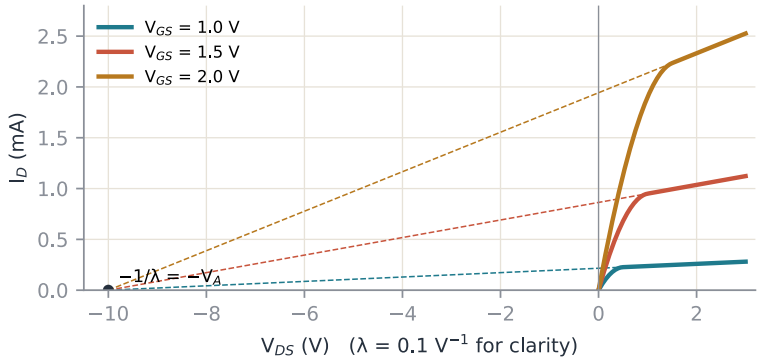


Fig 4.2 Saturation curves tilt upwards; extrapolations meet at −1/λ (exaggerated λ = 0.1 V^{-1}).

SOLVED EXAMPLE 4.3 · OUTPUT RESISTANCE

A MOSFET in saturation has $\lambda = 0.05\ V^{-1}$, and its ideal saturation current ($\lambda = 0$) is 1 mA. Find I_D at $V_{DS} = 1$ V and 2 V, and r_o.

▸ $I_D = 1(1 + 0.05) = 1.05$ mA at 1 V; 1.10 mA at 2 V.

▸ $r_o = 1/(\lambda I_D) \approx 1/(0.05 \times 10^{-3}) = 20$ kΩ. Doubling L roughly halves λ and doubles r_o (why analog designers use longer L).

1.05 mA, 1.10 mA; $r_o \approx 20$ kΩ

4.4 Drain-induced barrier lowering (DIBL)

In an off transistor, electrons in the source are held back by a potential barrier (the p-region under the gate). In a long device only the gate controls that barrier. In a short device the drain's depletion region reaches close to the source, and a high V_{DS} **pulls the barrier down**. More electrons get over it: the transistor leaks more, as if V_T had dropped.

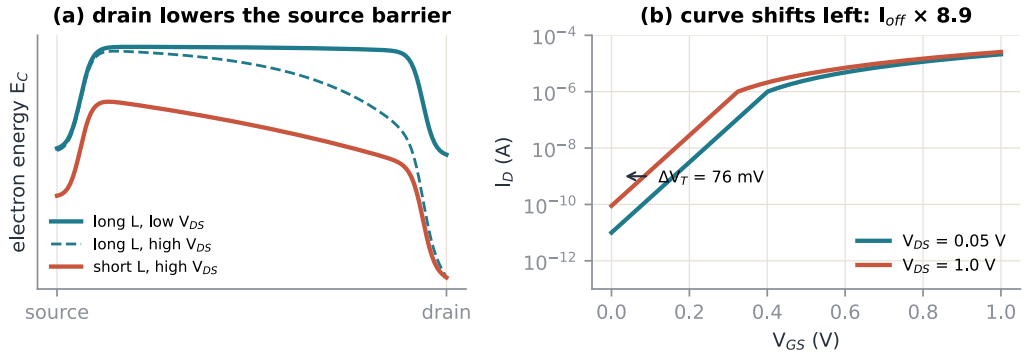


Fig 4.3 (a) Conduction-band edge along the channel (qualitative). (b) DIBL shifts the subthreshold curve left (Example 4.4).

DIBL

$$V_T = V_{T0} - \eta V_{DS} \qquad \eta = \text{DIBL coefficient (mV/V)} \qquad I_{off} \text{ rises by } 10^{\eta \Delta V_{DS}/S}$$

The extreme case is **punch-through**: source and drain depletion regions merge, the gate loses control completely and current flows deep in the substrate. Remedies: shallow junctions, halo (pocket) implants, thinner oxide, and ultimately multi-gate devices.

SOLVED EXAMPLE 4.4 · DIBL AND LEAKAGE GATE

A short-channel MOSFET has $\eta = 80$ mV/V and S = 80 mV/dec. By what factor does I_{off} grow when V_{DS} goes from 0.05 V to 1.0 V?

▸ $\Delta V_T = 80\ \text{mV/V} \times 0.95\ \text{V} = 76$ mV.

▸ Factor = $10^{76/80} = 10^{0.95} = 8.9$.

V_T drops 76 mV; off-current rises ≈ 8.9×

4.5 Velocity saturation

The square law assumes $v = \mu E$ with no upper limit. In silicon the electron velocity saturates at $v_{sat} \approx 10^7$ cm/s once the field exceeds a few 10^4 V/cm, because carriers lose their extra energy to the lattice (optical phonons). With L = 100 nm and V_{DS} = 1 V, the average lateral field is 10^5 V/cm, well into saturation.

VELOCITY-SATURATED MOSFET

$$v = \frac{\mu E}{1 + \mu E / v_{sat}} \qquad \text{Full velocity saturation: } I_{D,sat} = WC_{ox}(V_{GS} - V_T)v_{sat} \qquad g_m = WC_{ox}v_{sat}$$

Consequences: I_D becomes roughly **linear** in overdrive instead of quadratic, **independent of L** (shrinking L no longer buys current), g_m becomes independent of bias, and $V_{DS(sat)}$ is smaller than $V_{GS} - V_T$.

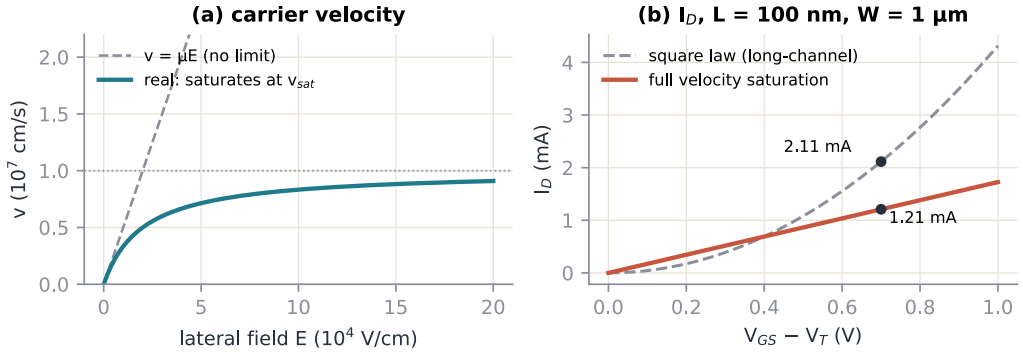


Fig 4.4 (a) Electron velocity vs field (μ = 500 cm²/V·s). (b) Square law vs velocity-saturated current for Example 4.5.

SOLVED EXAMPLE 4.5 · HOW WRONG IS THE SQUARE LAW?

An nMOS has W = 1 μm, L = 100 nm, t_{ox} = 2 nm, μ_n = 500 cm²/V·s, $v_{sat} = 10^7$ cm/s, $V_{GS} - V_T$ = 0.7 V. Compare $I_{D,sat}$ from the square law and from full velocity saturation. Find g_m in the latter case, and the L at which the two models give equal current.

▸ $C_{ox} = 3.453 \times 10^{-13} / 2 \times 10^{-7} = 1.727 \times 10^{-6}$ F/cm².

▸ Square law: ½ × 500 × 1.727×10^{-6} × 10 × 0.49 = 2.11 mA.

▸ Velocity saturation: $10^{-4} \times 1.727 \times 10^{-6} \times 0.7 \times 10^7$ = 1.21 mA. $g_m = WC_{ox}v_{sat}$ = 1.73 mS.

▸ Equal when $\tfrac{1}{2}\mu V_{ov}/L = v_{sat} \Rightarrow L = \mu V_{ov}/(2v_{sat}) = 500 \times 0.7/(2 \times 10^7) = 0.175$ μm. Below this, velocity saturation dominates.

Square law 2.11 mA vs 1.21 mA real (≈ 43 % lower); g_m = 1.73 mS; crossover L ≈ 0.18 µm

4.6 Hot-carrier effects

Near the drain of a short device the lateral field is huge. Electrons accelerated there gain far more energy than the thermal kT: they become **hot carriers**. Two bad things follow:

- **Impact ionisation:** hot electrons knock out new electron-hole pairs. The electrons go to the drain; the holes flow into the body as a **substrate current** I_{sub}, which can forward-bias the source junction and trigger latch-up or a parasitic bipolar action.
- **Injection into the oxide:** the hottest electrons jump the 3.1 eV Si/SiO_2 barrier and get trapped in the oxide or create interface states. Over months this shifts V_T, lowers g_m and mobility: a **reliability** (ageing) problem.

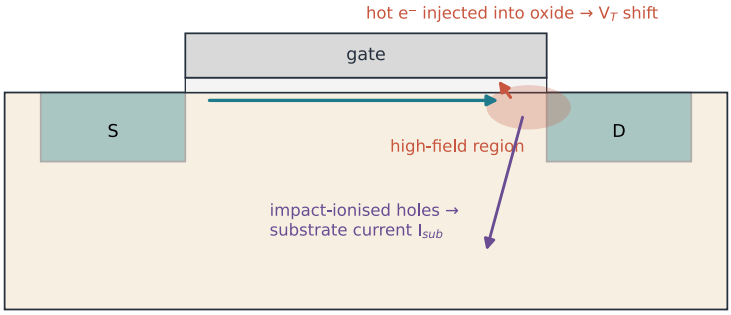


Fig 4.5 Hot-carrier mechanisms concentrate in the high-field region at the drain end of the channel.

Remedies: reduce the peak field with a **lightly doped drain (LDD)** extension, lower supply voltages, and graded junctions. The substrate current is used as a monitor for hot-carrier stress in reliability testing.

4.7 Threshold voltage roll-off

Long-channel theory says V_T does not depend on L. In short devices it does: V_T **falls** as L shrinks. Reason (**charge sharing**): part of the depletion charge under the gate actually belongs to the source and drain junction depletion regions, which the junctions "support" for free. The gate has to support less charge, so less gate voltage is needed. In a long device that shared part is a negligible fraction; in a short device it is a large fraction. DIBL adds further lowering at high V_{DS}.

CHARGE-SHARING MODEL (YAU), JUNCTION DEPTH R_J

$$\Delta V_T = -\frac{qN_a x_{dT}}{C_{ox}} \cdot \frac{r_j}{L}\left[\sqrt{(1 + 2x_{dT}/r_j)} - 1\right]$$

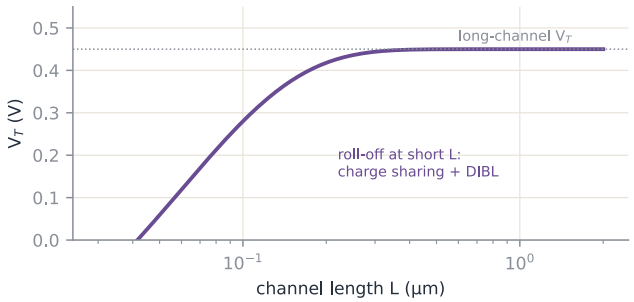


Fig 4.6 V_T roll-off (qualitative): flat for long channels, falling steeply below ≈ 0.2 µm.

Why it matters: every chip has a spread of gate lengths from lithography. Where the curve is steep, small L variations cause big V_T (and leakage) variations. Remedies: shallower junctions (small r_j), thinner oxide, halo implants (which can even produce **reverse** short-channel effect, V_T rising slightly before rolling off).

INTERVIEW CORNER

Q. List the short-channel effects.
A. V_T roll-off (charge sharing), DIBL, punch-through, channel length modulation, velocity saturation, mobility degradation, hot-carrier effects, and gate-induced drain leakage.

Q. Why can't we reduce V_T to 0.1 V to get more speed?
A. Subthreshold swing is at best 60 mV/dec, so with V_T = 0.1 V the current at V_{GS} = 0 would be less than two decades below its value at threshold. Multiplied over a billion transistors, the leakage power would be enormous.

Q. What does an LDD do?
A. A lightly doped extension between channel and n^+ drain spreads the voltage drop over a longer distance, lowering the peak field and so the hot-carrier damage, at the cost of some series resistance.

Q. Why did the industry move from planar to FinFET?
A. Planar devices below ≈ 25 nm could not keep DIBL, V_T roll-off and leakage under control. Wrapping the gate around a thin fin restores gate control, so leakage drops and V_{DD} can be lowered.

Q. Why is the FinFET channel left undoped?
A. Gate control comes from geometry, not doping; an undoped fin avoids random dopant fluctuation (V_T variability) and impurity scattering, so mobility improves. V_T is tuned by the gate metal work function.

Q. What comes after FinFET?
A. Gate-all-around nanosheet (or nanowire) FETs, with the gate on all four sides, then stacked complementary FETs (CFET).

4.8 The MESFET

The **ME**tal–**S**emiconductor FET replaces the MOS gate with a **Schottky barrier gate** (section 3.1). It is built on GaAs, which has no good native oxide (so no MOSFET), but has about 6× higher electron mobility than Si: ideal for microwave amplifiers and RF front ends.

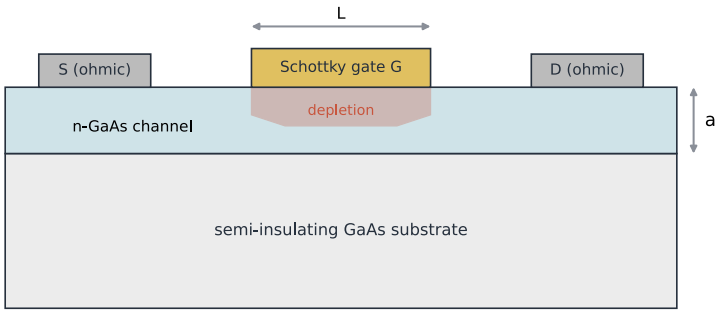


Fig 4.7 GaAs MESFET: a thin n-channel of thickness a on semi-insulating GaAs. The Schottky gate's depletion region sets how much channel is left.

Operation. Source and drain are ohmic contacts on a thin n-type channel. The Schottky gate's depletion region extends into the channel; making V_{GS} more negative widens it and narrows the conducting channel, like squeezing a hose. When the depletion region reaches the semi-insulating substrate, the channel is fully **pinched off**. As in a JFET, current is controlled by majority carriers and the gate draws almost no current (as long as the Schottky gate is not forward-biased beyond ≈ 0.7 V, which limits the gate swing compared with a MOSFET). Main uses: microwave low-noise and power amplifiers.

MESFET PINCH-OFF AND THRESHOLD

$$V_p = \frac{qN_d a^2}{2\varepsilon_s} \quad \text{(internal pinch-off voltage)} \qquad V_T = V_{bi} - V_p$$

$V_T < 0$: depletion-mode (normally ON, thick channel). $V_T > 0$: enhancement-mode (normally OFF, thin channel: the built-in depletion alone pinches it).

SOLVED EXAMPLE 4.6 · MESFET THRESHOLD

A GaAs MESFET ($\varepsilon_r = 13.1$) has $N_d = 10^{17}$ cm^{-3} and $V_{bi} = 0.8$ V. Find V_p and V_T for a = 0.2 µm and a = 0.1 µm. Which is enhancement-mode?

▸ $\varepsilon_s = 13.1 \times 8.854 \times 10^{-14} = 1.16 \times 10^{-12}$ F/cm.

▸ a = 0.2 µm: $V_p = 1.602 \times 10^{-19} \times 10^{17} \times (2 \times 10^{-5})^2/(2 \times 1.16 \times 10^{-12}) = 2.76$ V ⇒ $V_T = 0.8 - 2.76 = -1.96$ V.

▸ a = 0.1 µm: $V_p \propto a^2$ ⇒ 0.69 V ⇒ $V_T = +0.11$ V.

a = 0.2 µm: $V_T = -1.96$ V (depletion-mode) · a = 0.1 µm: $V_T = +0.11$ V (enhancement-mode)

4.9 The FinFET

Every short-channel effect has one root cause: the **drain competes with the gate** for control of the channel. A planar gate acts from the top only, leaving a deep leakage path it barely reaches. In a **FinFET** the channel is a thin vertical "fin" of silicon; the gate wraps over its top and both sides (tri-gate), and source and drain are the two ends of the fin.

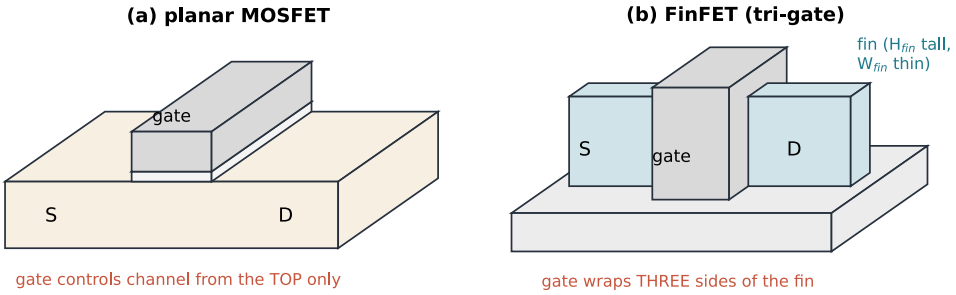


Fig 4.8 Planar MOSFET vs FinFET. In the FinFET the gate controls a thin fin from three sides.

Why it beats the planar MOSFET

- **Electrostatic control:** no point in the fin is far from the gate (fin width ≈ 5–10 nm), so the drain cannot reach under the gate. DIBL and V_T roll-off fall sharply; S approaches 60–70 mV/dec.
- **Lower leakage** at the same V_T, or a lower V_T and V_{DD} at the same leakage: big power savings.
- **Undoped (lightly doped) fin:** V_T is set by the gate work function instead of channel doping, so there are no random dopant fluctuations and mobility is higher.
- **More current per footprint:** the channel width runs up the fin sides, $W_{eff} = 2H_{fin} + W_{fin}$ per fin, and several fins can be packed at a tight pitch.

Trade-offs: width is quantised (you can only use 1, 2, 3… fins), fabrication is harder, and the fin heats up more (self-heating). Since Intel's 22 nm node (2011) FinFETs have been standard; below about 3 nm, **gate-all-around (GAA) nanosheets** wrap the gate around all four sides of stacked horizontal sheets for even better control.

SOLVED EXAMPLE 4.7 · FINFET EFFECTIVE WIDTH INTERVIEW

A FinFET has H_{fin} = 42 nm, W_{fin} = 8 nm, fin pitch 30 nm. Find W_{eff} per fin and for 3 fins. How does this compare with a planar device occupying the same 90 nm footprint?

- $W_{eff} = 2 \times 42 + 8 = 92$ nm per fin; 3 fins ⇒ 276 nm.
- 3 fins need $3 \times 30 = 90$ nm of layout width. A planar device in 90 nm has W = 90 nm.
- Ratio = 276/90 = 3.07: about 3× more drive width (current) in the same area.

92 nm per fin; 276 nm for 3 fins ≈ 3× the planar width in the same footprint

✓ Module 4: Quick revision

Constant-field scaling (κ): dimensions, V ÷κ; N ×κ; I ÷κ; delay ÷κ; P ÷κ^2; density ×κ^2; power density constant.

Constant-voltage: field ×κ; I ×κ; delay ÷κ^2; power density ×κ^3; reliability problems.

Subthreshold: $I_D \propto e^{(V_{GS}-V_T)/nV_T}$, $n = 1 + C_{dep}/C_{ox}$; diffusion current.

Swing: $S = 2.3(kT/q)n \geq 59.6$ mV/dec at 300 K.

CLM: $I_D(1 + \lambda V_{DS})$; $r_o = 1/\lambda I_D$; $\lambda \propto 1/L$.

DIBL: $V_T = V_{T0} - \eta V_{DS}$; $I_{off} \times 10^{\eta \Delta V/S}$; extreme: punch-through.

Velocity saturation: $v_{sat} \approx 10^7$ cm/s; $I_D = WC_{ox}V_{ov}v_{sat}$ (linear, L-independent); $g_m = WC_{ox}v_{sat}$.

Hot carriers: impact ionisation (I_{sub}), oxide injection (V_T drift); fix with LDD.

V_T roll-off: charge sharing with S/D junctions; worse for deep junctions, thick oxide.

MESFET: Schottky gate on GaAs; $V_p = qN_da^2/2\varepsilon_s$; $V_T = V_{bi} - V_p$.

FinFET: gate on 3 sides of thin fin; $W_{eff} = 2H_{fin} + W_{fin}$; low DIBL, near-ideal S, undoped channel. Next: GAA nanosheets.

Practice: short questions (3 marks)

1. Why does power density stay constant in constant-field scaling?
2. Define subthreshold swing. Why can it not go below 60 mV/dec at 300 K?
3. Find S for $C_{dep}/C_{ox} = 0.5$. [89.5 mV/dec]
4. What is channel length modulation? Give its effect on r_o.
5. Explain DIBL with a band sketch.
6. What are hot carriers? Name two effects.
7. Why is GaAs preferred for MESFETs?
8. Give three advantages of FinFET over planar MOSFET.

Practice: long questions (9/14 marks)

1. Compare constant-field and constant-voltage scaling (table of parameters).
2. Explain subthreshold conduction and derive S. A device has S = 90 mV/dec; how much must V_T rise to cut I_{off} by 100×? [180 mV]
3. Explain DIBL, V_T roll-off and punch-through, with remedies.
4. Derive $I_{D,sat}$ under full velocity saturation. Find it for W = 2 µm, t_{ox} = 3 nm, V_{ov} = 0.5 V. [1.15 mA]
5. Explain GaAs MESFET structure and operation; depletion vs enhancement mode.
6. Explain the FinFET structure and why it suppresses short-channel effects.

⚡ Placement & GATE rapid-fire

Cover the answer, say yours out loud, then check. The ones marked ★ are the "tricky" conceptual questions interviewers use to separate memorisers from understanders.

Q. What is a hole?
A. An empty valence-band state that moves like a particle with charge +q and its own effective mass.

Q. Why is Si used over Ge for most ICs?
A. Larger E_g (lower leakage, higher operating temperature) and an excellent native oxide, SiO_2.

Q. Where is E_F in intrinsic Si?
A. Almost at mid-gap, 12.8 meV below it, because $m_n^* > m_p^*$.

Q. ★ Doping Si with 10^{16} donors: how does n_i change?
A. It does not. n_i is a material property (E_g, T). Only n_0 and p_0 change, with $n_0 p_0 = n_i^2$.

Q. How much does E_F move per decade of doping?
A. kT ln 10 = 59.6 meV at 300 K.

Q. What happens to an n-type sample at very high T?
A. n_i exceeds N_d; it becomes intrinsic and E_F returns to E_i.

Q. Two scattering mechanisms that limit mobility?
A. Lattice (phonon) scattering, worse at high T; ionised impurity scattering, worse at high doping.

Q. State the Einstein relation.
A. $D/\mu = kT/q \approx 25.9$ mV at 300 K.

Q. ★ The Hall voltage of a sample is zero. What can you conclude?
A. Electron and hole contributions cancel ($n\mu_n^2 = p\mu_p^2$): a nearly intrinsic or specially compensated sample.

Q. Lifetime vs diffusion length?
A. τ is survival time; $L = \sqrt{(D\tau)}$ is the distance covered in that time.

Q. What does the continuity equation express?
A. Conservation of carriers: change = net inflow + generation − recombination.

Q. What do quasi-Fermi levels measure?
A. Non-equilibrium carrier densities; their split $E_{Fn} - E_{Fp}$ sets $np = n_i^2 e^{(E_{Fn}-E_{Fp})/kT}$.

Q. Typical V_{bi} of a Si junction?
A. 0.6–0.9 V, depending on doping (0.754 V for $10^{17}/10^{16}$).

Q. ★ Why can't you measure V_{bi} with a voltmeter?
A. Metal-semiconductor contact potentials around the loop cancel it exactly; equilibrium means no net EMF.

Q. How does W depend on reverse bias?
A. $W \propto \sqrt{(V_{bi} + V_R)}$ for an abrupt junction.

Q. Which side of a junction dominates the diode current?
A. The lightly doped side, because its minority concentration n_i^2/N is larger.

Q. ★ Diode current at fixed V roughly doubles every few degrees. Why?
A. $I_s \propto n_i^2 \propto T^3 e^{-E_g/kT}$; at fixed current V_D falls ≈ 2 mV/°C.

Q. What is the ideality factor?
A. η in $e^{V/\eta V_T}$: 1 for pure diffusion current, → 2 when depletion-region recombination dominates.

Q. Diffusion vs junction capacitance?
A. C_d: stored minority charge, $\propto I_D\tau$, forward bias. C_j: depletion charge, $\propto (V_{bi} + V_R)^{-1/2}$, reverse bias.

Q. What limits diode switching speed?
A. Removal of stored minority charge (reverse recovery), set by lifetime.

Q. Zener vs avalanche: temperature coefficient?
A. Zener negative, avalanche positive; they cancel near 5–6 V.

Q. How do you get high β?
A. $N_E \gg N_B$ ($\gamma \to 1$) and $W_B \ll L_B$ ($\alpha_T \to 1$).

Q. ★ α changes from 0.99 to 0.995. What happens to β?
A. It doubles, 99 to 199: $\beta = \alpha/(1-\alpha)$ magnifies tiny changes in α.

Q. Shape of the base minority profile in active mode?
A. Almost a straight line from $n_{B0}e^{V_{BE}/V_T}$ to ≈ 0; its slope is I_C.

Q. Early voltage, physically?
A. Base-width modulation; V_A sets $r_o = V_A/I_C$.

Q. Schottky barrier height, ideal formula?
A. $\varphi_{B0} = \varphi_m - \chi$ (n-type); in practice pinned by interface states.

Q. Why is a Schottky diode fast?
A. Majority-carrier device: no stored minority charge, so no reverse recovery.

Q. How do you make an ohmic contact?
A. Heavy doping under the metal, so the barrier is thin enough to tunnel through.

Q. ★ The HF C-V curve of a MOS capacitor shifts left. What happened?
A. More positive fixed oxide charge (or a lower φ_{ms}): V_{FB} became more negative. Shape is unchanged.

Q. Condition for strong inversion?
A. $\psi_s = 2\varphi_{fp}$: surface electron density equals bulk hole density.

Q. Three components of V_T?
A. V_{FB} (work function + oxide charge), $2\varphi_{fp}$ (band bending), Q_{SD}/C_{ox} (depletion charge).

Q. Effect of thinner oxide on V_T (nMOS)?
A. Larger C_{ox} shrinks the Q_{SD}/C_{ox} and Q_{ss}/C_{ox} terms, so V_T moves towards $\varphi_{ms} + 2\varphi_{fp}$, and the body effect weakens.

Q. What is the body effect?
A. V_T rises with V_{SB}: $\Delta V_T = \gamma(\sqrt{(2\varphi_f + V_{SB})} - \sqrt{(2\varphi_f)})$.

Q. ★ In saturation the channel is pinched off. How does current flow?
A. Electrons reaching the pinch-off point are swept through the depleted region by the high lateral field.

Q. g_m of a MOSFET in saturation?
A. $2I_D/V_{ov} = \sqrt{(2k'(W/L)I_D)}$; compare I_C/V_T for a BJT.

Q. Why is NMOS faster than PMOS?
A. $\mu_n \approx 2$–$3\ \mu_p$, so more current per W/L.

Q. What is Dennard scaling and why did it end?
A. Constant-field scaling (power density constant). It ended because V_T, and hence V_{DD}, could not keep scaling due to subthreshold leakage.

Q. Minimum subthreshold swing at 300 K?
A. 2.3kT/q ≈ 60 mV/decade (n = 1).

Q. ★ How can a device beat 60 mV/dec?
A. Not with thermionic injection; it needs a different mechanism, such as band-to-band tunnelling (TFET) or negative-capacitance FETs.

Q. DIBL in one line?
A. High V_{DS} lowers the source barrier in short channels, reducing V_T and raising leakage.

Q. Effect of velocity saturation on I_D?
A. $I_D \propto V_{ov}$ instead of V_{ov}^2, and independent of L.

Q. What is an LDD and why?
A. Lightly doped drain extension that lowers the peak field to limit hot-carrier damage.

Q. Why is the MESFET built in GaAs?
A. GaAs has high electron mobility but no good native oxide, so a Schottky gate replaces the MOS gate.

Q. FinFET effective width?
A. $2H_{fin} + W_{fin}$ per fin, times the number of fins (quantised).

Q. ★ Why not just make planar oxide thinner instead of going FinFET?
A. Below ≈ 1.2 nm SiO_2, gate tunnelling leakage explodes, and the deep substrate leakage path remains; the fin removes that path geometrically.

∑ Formula sheet

Module 1: Semiconductor physics

Quantity	Formula
Fermi–Dirac	$f(E) = 1/[1 + e^{(E - E_F)/kT}]$
Equilibrium concentrations	$n_0 = N_c e^{-(E_C - E_F)/kT} = n_i e^{(E_F - E_i)/kT}$; $p_0 = N_v e^{-(E_F - E_V)/kT} = n_i e^{(E_i - E_F)/kT}$
Mass action; n_i	$n_0 p_0 = n_i^2 = N_c N_v e^{-E_g/kT}$; $n_i \propto T^{3/2} e^{-E_g/2kT}$
Charge neutrality	$n_0 = (N_d - N_a)/2 + \sqrt{[((N_d - N_a)/2)^2 + n_i^2]}$
Intrinsic level	$E_i - E_{mid} = ¾kT \ln(m_p^*/m_n^*)$
Drift; conductivity	$J = q(n\mu_n + p\mu_p)E$; $\sigma = q(n\mu_n + p\mu_p)$; $\mu = q\tau_c/m^*$
Hall effect	$V_H = IB/(qpd)$; $R_H = 1/qp$ $(-1/qn)$; $\mu = I\,L/(qpV_x Wd) = \sigma\lvert R_H\rvert$
Diffusion; Einstein	$J_n = qD_n dn/dx$, $J_p = -qD_p dp/dx$; $D/\mu = kT/q$
Recombination	$R = \delta p/\tau_p$; $\delta p = G\tau_p$; $\delta p(t) = \delta p(0) e^{-t/\tau}$
Poisson; continuity	$dE/dx = \rho/\varepsilon_s$; $\partial p/\partial t = -(1/q)\partial J_p/\partial x + G - R$
Diffusion length	$L = \sqrt{(D\tau)}$; $\delta p(x) = \delta p(0) e^{-x/L}$; $J(0) = qD\delta p(0)/L$
Quasi-Fermi levels	$n = n_i e^{(E_{Fn} - E_i)/kT}$, $p = n_i e^{(E_i - E_{Fp})/kT}$

Module 2: PN junction and BJT

Quantity	Formula
Built-in potential	$V_{bi} = V_T \ln(N_a N_d/n_i^2)$
Depletion width	$W = \sqrt{[2\varepsilon_s(V_{bi} + V_R)(N_a + N_d)/(qN_a N_d)]}$; $x_n = WN_a/(N_a + N_d)$
Peak field	$E_{max} = qN_d x_n/\varepsilon_s = 2(V_{bi} + V_R)/W$
Law of the junction	$p_n(x_n) = p_{n0} e^{V_a/V_T}$
Ideal diode	$I = I_s(e^{V/V_T} - 1)$; $I_s = Aqn_i^2(D_n/L_n N_a + D_p/L_p N_d)$
Capacitances	$C_j = \varepsilon_s A/W$; $C_d = \tau I_D/2V_T$ (long); $g_d = I_D/V_T$
Breakdown	$V_B \approx \varepsilon_s E_{crit}^2/(2qN_d)$
BJT gain	$\alpha = \gamma\alpha_T\delta$; $\beta = \alpha/(1 - \alpha)$; $\gamma \approx [1 + N_B D_E W_B/(N_E D_B L_E)]^{-1}$; $\alpha_T \approx 1 - W_B^2/2L_B^2$
Early effect	$I_C = I_S e^{V_{BE}/V_T}(1 + V_{CE}/V_A)$; $r_o = V_A/I_C$

Module 3: M-S contacts, MOS, MOSFET

Quantity	Formula
Schottky barrier (n)	$\varphi_{B0} = \varphi_m - \chi$; $V_{bi} = \varphi_{B0} - V_T \ln(N_c/N_d)$; $J_{sT} = A^*T^2 e^{-\varphi_B/V_T}$
MOS basics	$C_{ox} = \varepsilon_{ox}/t_{ox}$; $\varphi_{fp} = V_T \ln(N_a/n_i)$; $x_{dT} = \sqrt{(4\varepsilon_s \varphi_{fp}/qN_a)}$
C-V	$C_{min} = \varepsilon_{ox}/[t_{ox} + (\varepsilon_{ox}/\varepsilon_s)x_{dT}]$
Flat band	$V_{FB} = \varphi_{ms} - Q'_{ss}/C_{ox}$; n^+ poly/p-Si: $\varphi_{ms} = -(E_g/2q + \varphi_{fp})$
Threshold	$V_{TN} = V_{FB} + 2\varphi_{fp} + \sqrt{(4q\varepsilon_s N_a \varphi_{fp})}/C_{ox}$
Body effect	$\Delta V_T = \gamma[\sqrt{(2\varphi_{fp} + V_{SB})} - \sqrt{(2\varphi_{fp})}]$; $\gamma = \sqrt{(2q\varepsilon_s N_a)}/C_{ox}$
MOSFET I-V	Linear: $k'(W/L)[V_{ov}V_{DS} - V_{DS}^2/2]$; sat: $\frac{1}{2}k'(W/L)V_{ov}^2$; $k' = \mu C_{ox}$
Transconductance	$g_m = k'(W/L)V_{ov} = 2I_D/V_{ov} = \sqrt{(2k'(W/L)I_D)}$

Module 4: Scaling and advanced devices

Quantity	Formula
Constant-field scaling	L, V ÷ κ, N × κ ⇒ I ÷ κ, delay ÷ κ, P ÷ κ^2, power density constant
Subthreshold	$I_D \propto e^{(V_{GS}-V_T)/nV_T}$; $S = 2.3(kT/q)(1 + C_{dep}/C_{ox})$
CLM; DIBL	$I_D(1 + \lambda V_{DS})$, $r_o = 1/\lambda I_D$; $V_T = V_{T0} - \eta V_{DS}$
Velocity saturation	$I_{D,sat} = WC_{ox}V_{ov}v_{sat}$; $g_m = WC_{ox}v_{sat}$
V_T roll-off (Yau)	$\Delta V_T = -(qN_a x_{dT}/C_{ox})(r_j/L)[\sqrt{(1 + 2x_{dT}/r_j)} - 1]$
MESFET	$V_p = qN_d a^2/2\varepsilon_s$; $V_T = V_{bi} - V_p$
FinFET	$W_{eff} = n_{fin}(2H_{fin} + W_{fin})$

☾ Night before the exam

Breathe. You do not need to re-read the book. Read this page, sketch the five diagrams below once each from memory, and sleep.

DRAW THESE FIVE FROM MEMORY

1. Band diagram of a PN junction: equilibrium, forward, reverse (E_F flat → splits by qV).
2. Charge, field and potential profiles of an abrupt junction (rectangles → triangle → parabola).
3. Minority carrier profiles in a forward-biased diode and in a BJT base (exponential tails; straight line in the base).
4. MOS band diagrams in accumulation, depletion, inversion, plus the LF/HF C-V curve.
5. MOSFET output characteristics with the $V_{DS} = V_{GS} - V_T$ boundary, and log I_D-V_{GS} showing S.

TEN FACTS WORTH MARKS

- $n_0p_0 = n_i^2$, $n_i = 1.5 \times 10^{10}$ cm^{-3}.
- $E_F - E_i = kT \ln(N/n_i)$.
- $D/\mu = kT/q$; $L = \sqrt{(D\tau)}$.
- $V_H = IB/(qnd)$; sign gives type.
- $V_{bi} = V_T \ln(N_aN_d/n_i^2)$; $W \propto \sqrt{(V_{bi} + V_R)}$.
- $I = I_s(e^{V/V_T} - 1)$; 60 mV/decade.
- $\beta = \alpha/(1 - \alpha)$; thin base, heavy emitter.
- $V_T = V_{FB} + 2\varphi_{fp} + Q_{SD}/C_{ox}$.
- $I_{D,sat} = \frac{1}{2}\mu C_{ox}(W/L)V_{ov}^2$; $g_m = 2I_D/V_{ov}$.
- $S \geq 60$ mV/dec; FinFET $W_{eff} = 2H + W$.

TEN WAYS MARKS ARE LOST

- Mixing cm and m (Hall problems: use SI).
- Using W instead of d in V_H.
- Wrong sign of electron diffusion current.
- Forgetting n_i^2/N for minority carriers.
- Not stating diode assumptions.
- Depletion on the wrong (heavy) side.
- Confusing MOSFET saturation with BJT saturation.
- Not checking the region (V_{DS} vs V_{ov}) first.
- Sign of φ_{ms} for n^+ vs p^+ poly.
- Calling V_T roll-off "DIBL" (related, not identical).

ONE-LINE STORY OF THE WHOLE SUBJECT

Doping decides where E_F sits; joining materials forces E_F to line up, which bends the bands; bias splits E_F and lets minority carriers flow (diode, BJT) or lets a gate pull a channel to the surface (MOSFET); and shrinking that MOSFET is a constant fight to keep the gate, not the drain, in charge.